بسم الله الرحمن الرحيم

University of Khartoum

Faculty of Engineering

Mechanical Engineering Department

Techno-Economic Feasibility of Solar-Powered Irrigation Systems

A Case Study in River Nile State, Sudan

A thesis submitted in partial fulfillment of the requirements for the degree of B.Sc. in Mechanical Engineering

Submitted by:

1/ Hisham Salih Hassan Salih 2/ Mohamed Hatim Fadul Hassan

Supervised by:

Dr. Salah Ahmed Abdalla

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Dedication

We dedicate this work to our dear parents and families, whose love, support, and prayers have guided us through every step.

To our close friends, who shared this long journey with us at the University of Khartoum, thank you for being there through the challenges and the good times.

This achievement wouldn’t have been the same without you.

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Abstract

This study evaluates the technical and economic feasibility of replacing diesel-powered irrigation systems with solar-powered alternatives for smallholder wheat farming in Abu Hamed, River Nile State, Sudan. In response to high fuel costs, unreliable diesel supply chains, and environmental pressures, solar energy offers a promising solution for off-grid agricultural irrigation in arid regions.

The research combines agro-climatic and energy modeling tools. The Food and Agriculture Organization's **CROPWAT** model is used to determine crop water requirements based on local climatic data, while **HOMER Pro** is employed to simulate and compare diesel and solar-powered pumping systems. Key indicators such as seasonal irrigation demand, load profiles, energy costs, and emissions are assessed to support system design and decision-making.

The analysis is based on a **10-hectare** wheat farm, with a solar system sized at approximately **6.4 kW** to meet irrigation needs. **Results show that the solar-powered system reduces operating costs by around 60% and eliminates more than 3.5 tons of CO_2 emissions per season**. The study also highlights key barriers to adoption, including financing constraints, limited technical awareness, and the absence of localized planning frameworks.

By integrating water demand modeling with energy system optimization for a defined crop and location, this research provides a practical, data-driven evaluation of solar irrigation under Sudanese conditions. It contributes to ongoing development efforts aimed at enhancing agricultural resilience and energy access through renewable technologies, while offering a replicable approach for similar rural contexts.

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Chapter One

Introduction

1 Introduction

1.1 Background and Context

Ensuring reliable access to both water and energy is fundamental to agricultural development, especially in arid and semi-arid regions. In Sudan, where agriculture forms the backbone of rural livelihoods and food security, farmers increasingly depend on irrigation to cope with low and irregular rainfall. However, in many regions such as the River Nile State, the infrastructure needed to support irrigation remains weak. Smallholder farmers often rely on diesel-powered pumps, which, despite their widespread use, present growing challenges due to fuel scarcity, fluctuating prices, and operational inefficiencies.

As energy costs rise and diesel supply chains become less reliable, solar-powered irrigation systems are gaining attention as a promising alternative. Sudan has excellent solar potential, and solar photovoltaic (PV) systems can provide a decentralized and clean energy source, especially in off-grid areas. These systems are already being used in similar regions across sub-Saharan Africa and Asia to lower irrigation costs, improve resilience, and reduce emissions. However, despite the apparent benefits and increasing international interest, the adoption of solar irrigation in Sudan remains limited.

This study focuses on Abu Hamed, a rural area in River Nile State with high solar irradiance, direct access to surface water from the Nile, and a strong reliance on diesel-powered irrigation. These features make it a representative location for evaluating the viability of solar irrigation in Sudan's agricultural transition.

1.2 Relevance and Research Gap

Sudan's agricultural sector continues to rely heavily on diesel-powered irrigation, particularly in remote areas like Abu Hamed in the River Nile State. However, rising fuel prices, unstable supply chains, and limited maintenance infrastructure are making diesel systems increasingly unsustainable. These challenges directly threaten the productivity and financial viability of smallholder farmers, who already operate under tight resource constraints.

At the same time, Sudan possesses exceptional solar energy potential, and the falling cost of photovoltaic technology offers an opportunity to transition toward more reliable and

climate-friendly irrigation systems. Initiatives are already underway to introduce solar pumping in parts of the country, indicating institutional support and policy momentum. Yet despite these developments, the adoption of solar irrigation remains limited, and clear guidance is lacking on how to effectively implement such systems under local conditions.

While the global literature highlights the technical feasibility and long-term benefits of solar irrigation, most existing studies remain broad in scope. Few address the specific operational realities of small farms or incorporate crop-level water requirements into energy system design. Prior research often uses simplified assumptions or generalized load estimates, which limits their practical relevance in planning and implementation.

This study addresses that gap by integrating crop water modeling with energy simulation to evaluate whether solar-powered systems can realistically replace diesel irrigation for wheat farming in Abu Hamed. By aligning irrigation demand with energy supply under local climatic conditions, it offers a more grounded assessment of technical feasibility, cost-effectiveness, and long-term sustainability. The approach aims to contribute both to applied research and to ongoing development efforts seeking to enhance agricultural resilience through renewable energy.

1.3 Research Objectives and Questions

Building on the identified research gap, this study assesses the feasibility of solar-powered irrigation using localized crop water data and economic modeling tailored to the Abu Hamed context. The goal is to deliver a data-grounded alternative to existing diesel systems, reflecting real operational conditions faced by smallholder farmers. It focuses on a representative small-scale wheat farm and employs agro-climatic modeling (CROPWAT and CLIMWAT) and techno-economic simulation (HOMER Pro) to evaluate water demand, energy requirements, and system costs. Wheat was selected due to its national importance as a staple crop, its widespread cultivation in Sudan, and its relatively high and seasonally concentrated irrigation demand, which makes it suitable for load-based energy system modeling. By doing so, it offers practical insights into the conditions under which solar irrigation can be adopted sustainably in rural Sudan.

The research is guided by the following questions:

- How does the technical and economic performance of solar-powered irrigation compare to diesel-powered alternatives in the River Nile State?
- What are the key environmental and operational implications of adopting solar over diesel irrigation?
- What constraints must be addressed to make solar irrigation systems viable for smallholder farmers?

1.4 Scope and Limitations

This study focuses on assessing the technical and economic feasibility of transitioning from diesel to solar-powered irrigation for a smallholder wheat farm located in Abu Hamed, River Nile State. The scope is limited to a single 10-hectare farm and one crop type, with simulations based on local agro-climatic data and hourly load estimates.

Irrigation demand is calculated using long-term average climatic inputs via the FAO's CROPWAT model, while system performance and cost analysis are conducted using HOMER Pro, based on technology and price data representative of conditions in 2025. No battery storage is modeled, and the analysis assumes a steady daytime irrigation profile, consistent with typical surface irrigation practices in the region.

The study does not include behavioral or institutional dimensions such as user training, maintenance capacity, or policy incentives. Additionally, site-specific field measurements were not possible, and reliance on modeling tools introduces some abstraction from real-world operational variability. Nevertheless, the approach is designed to be replicable and adaptable to similar agricultural contexts across arid and semi-arid regions in Sudan.

The structure of the thesis is as follows:

Chapter 2 reviews the literature on solar water pumping and renewable energy in agriculture, with attention to global trends and Sudan's specific context.

Chapter 3 explains the methodological framework, tools, and data sources.

Chapter 4 presents the case study of Abu Hamed, including its climate, soil, and agricultural relevance.

Chapter 5 discusses the results of the crop water requirement analysis, energy simulations, and economic comparisons.

Chapter 6 concludes the study with a summary of key findings, implications for practice, and recommendations for future research.

Chapter Two

Literature Review

2 Literature Review

2.1 Introduction

In Sudan's rural agricultural landscape, small-scale farmers face mounting challenges in sustaining irrigation due to water scarcity, unreliable energy access, and rising operational costs. The traditional reliance on diesel-powered water pumping systems, while familiar and relatively inexpensive to install, has become increasingly unsustainable amid fluctuating fuel prices, maintenance difficulties, and frequent supply chain disruptions. These issues are especially pronounced in off-grid regions such as the River Nile State, where electrification rates remain among the lowest in the country, and the agricultural sector is heavily dependent on costly and environmentally harmful diesel generators.(E. Ahmed, 2020)

Given Sudan's high solar irradiance, particularly in its northern and central regions, solar-powered irrigation has emerged as a promising alternative. These systems harness photovoltaic (PV) technology to deliver water for irrigation without fuel dependency, offering a renewable and decentralized energy solution tailored to Sudan's climatic and infrastructural realities. However, the adoption of solar pumping systems is constrained by a range of barriers, including high upfront investment costs, lack of access to financing, and limited technical knowledge among farmers. For many rural households, the transition to solar irrigation remains aspirational without significant external financial or institutional support.(N. Ibrahim, 2024)

This literature review builds a conceptual foundation for assessing the feasibility and broader implications of solar water pumping in Sudan, particularly for small-scale farmers operating in the River Nile State. It examines the evolution of solar irrigation technologies and their global and regional diffusion, compares solar and diesel systems across economic and environmental dimensions, and explores the institutional and socioeconomic enablers and constraints that shape technology adoption.

The analysis focuses on addressing critical gaps in existing knowledge by assessing the technical feasibility, economic implications, and environmental benefits of solar-powered irrigation under real-world constraints.

2.2 Solar Water Pumping

Solar water pumping refers to the use of solar-powered systems to drive water pumps for agricultural, domestic, or industrial purposes, primarily focusing on irrigation in regions with limited energy infrastructure. These systems typically consist of solar photovoltaic (PV) panels that convert sunlight into electricity, powering an electric motor to pump water from underground wells, surface water sources, or reservoirs. (Miller et al., 2024)

Solar water pumping offers an integrated solution to the dual challenges of water and energy scarcity. By providing a decentralized and low-cost energy source, solar-powered pumping can enable more reliable irrigation scheduling, improve water use efficiency, and reduce environmental impacts. In regions with high solar potential and limited grid infrastructure, such as the River Nile State, these systems align with both energy access goals and agricultural development priorities. However, while the technical feasibility of solar irrigation has been demonstrated in various settings globally, its large-scale adoption in Sudan requires a closer examination of local conditions, financial barriers, and the institutional landscape.(Aliyu et al., 2018)

The adoption of solar water pumping has steadily increased as technological advancements and cost reductions have made these systems more accessible globally. This global shift towards sustainable irrigation practices highlights the growing recognition of solar water pumping as a practical solution for regions facing significant agricultural challenges. The following section delves into the global context of solar water pumping, examining its role in improving food security, addressing water shortages, and promoting sustainable development in diverse agricultural landscapes.(Push & Shalom, 2023)

2.2.1 Global Context of Solar Water Pumping

Solar water pumping systems have emerged as a cornerstone of sustainable agricultural practices worldwide. These systems harness solar photovoltaic (PV) technology to provide a reliable water supply for irrigation, addressing critical challenges related to water scarcity and energy access. Their widespread adoption is driven by advancements in solar technology, falling costs, and the need to transition from fossil fuels to renewable energy sources.(Push & Shalom, 2023)

Adoption Across Developed and Developing Nations

In developed countries, solar water pumping is often integrated into advanced agricultural systems, enhancing resource efficiency and reducing environmental impact. For instance, in Australia and the United States, solar-powered pumps are commonly used in large-scale irrigation projects to optimize water distribution while cutting carbon emissions. These systems often incorporate smart grid technologies and automation to further improve efficiency and resilience.(Patay & Montvajszki, 2015)

In developing countries, solar water pumping has become a transformative solution for smallholder farmers in off-grid areas. India's PM-KUSUM program exemplifies large-scale implementation in this context. Launched in 2019, the program aims to deploy millions of solar pumps across rural areas, reducing diesel dependency and providing farmers with consistent irrigation solutions. The program not only improves water access but also enables farmers to sell surplus electricity to the grid, creating additional income streams. Similar initiatives in Bangladesh and Vietnam have demonstrated the potential for solar water pumping to empower rural communities and enhance food security.(Kanna et al., 2020)

Environmental and Economic Impact

The environmental benefits of solar water pumping are significant. By replacing diesel-powered systems, solar pumps contribute to substantial reductions in greenhouse gas emissions. For instance, studies indicate that solar pumps can eliminate up to 70% of the emissions associated with diesel irrigation systems, aligning with global climate action goals. Additionally, solar water pumping supports sustainable water management by enabling precise irrigation schedules that reduce water wastage.(Guno & Agaton, 2022)

Economically, solar water pumping systems provide long-term savings for farmers. While the initial investment is higher compared to diesel systems, the absence of fuel costs and lower maintenance requirements allow these systems to pay for themselves within three to five years. Reports from the International Renewable Energy Agency (IRENA) highlight that, farmers transitioning to solar pumps save an average of $500 to $1,200 annually, making the technology both practical and economically viable.(Guno & Agaton, 2022)

Supporting Global Sustainability Goals

Solar water pumping directly contributes to international frameworks like the United Nations Sustainable Development Goals (SDGs). It supports SDG 7 (Affordable and Clean Energy) by providing renewable energy access, SDG 2 (Zero Hunger) by enhancing agricultural productivity, and SDG 13 (Climate Action) by reducing carbon emissions. These systems have become integral to global efforts to address energy poverty and ensure sustainable development.(Falchetta et al., 2023)

Figure 2.1: United Nation's Sustainable Development Goals

This figure highlights key global development priorities; Goals 2, 7, and 13 are especially relevant to solar-powered irrigation in terms of food security, clean energy access, and climate action.

2.2.2 Regional Context of Solar Water Pumping

Africa stands out as a region with immense potential for solar water pumping due to its abundant solar resources and pressing energy and water challenges. With over 600 million people lacking access to electricity and a significant portion of the agricultural sector dependent on unreliable or costly energy sources, the need for decentralized and sustainable irrigation solutions is critical. Solar water pumping systems have emerged as a transformative technology to address these interconnected challenges, particularly in rural and off-grid areas.(Falchetta et al., 2023)

Energy and Water Challenges in Africa

Energy access in Africa remains one of the lowest globally, with electrification rates in rural areas often below 20%. This limited energy infrastructure severely impacts agricultural productivity, as farmers in remote areas rely heavily on diesel-powered pumps for irrigation. These systems are not only costly to operate but also contribute to environmental degradation through greenhouse gas emissions. Additionally, water scarcity, exacerbated by climate change and erratic rainfall patterns, creates further barriers to sustainable agriculture.(Falchetta et al., 2023)

Key Initiatives in Africa

Several countries in Africa have implemented solar water pumping projects to address these challenges. In Kenya, programs such as the Renewable Energy and Adaptation to Climate Technologies (REACT) initiative have facilitated the deployment of thousands of solar water pumps. These systems have improved irrigation access, reduced reliance on diesel, and strengthened resilience to climate variability. Similarly, Ethiopia has integrated solar water pumping into its renewable energy strategies, irrigating thousands of hectares and improving food security in water-scarce regions.(Cowell et al., 2022)

Nigeria has also embraced solar water pumping, particularly in arid and semi-arid areas where conventional water sources are unreliable. These projects, often supported by development agencies, have enhanced water availability for agriculture while reducing the financial burden on smallholder farmers. Such initiatives underscore the adaptability of solar water pumping systems in meeting the unique needs of African communities.(Muhammadu, 2014)

Africa’s experience with solar water pumping offers valuable insights for Sudan, where similar energy and water challenges persist. The proven success of projects across the continent demonstrates the potential for tailored solutions that address local conditions.

2.3 Renewable Energy in Sudan's Agricultural Sector

2.3.1 Sudan energy Sector

Sudan's energy sector is characterized by its reliance on traditional energy sources, a centralized but limited grid infrastructure, and untapped renewable energy potential. Spanning a vast geographical area of over 1.8 million square kilometers, Sudan is endowed with abundant natural resources, including significant solar and wind energy potential, as well as hydropower from the Nile and its tributaries. Despite this, the country faces persistent challenges in meeting its energy demands, with significant disparities in access between urban and rural areas.(N. Ibrahim, 2024)

Biomass is the dominant source of energy in Sudan (illustrated in Figure 2.2), primarily utilized for domestic heating and cooking, while fossil fuels play a significant role in powering transportation, industry, and small-scale diesel generators. Hydropower, despite the presence of major dams such as the Merowe and Roseires Dams, makes a minimal contribution to the energy mix. The reliance on biomass and fossil fuels highlights the country's limited infrastructure for modern energy solutions and contributes to environmental issues such as deforestation and greenhouse gas emissions. (N. Ibrahim, 2024)

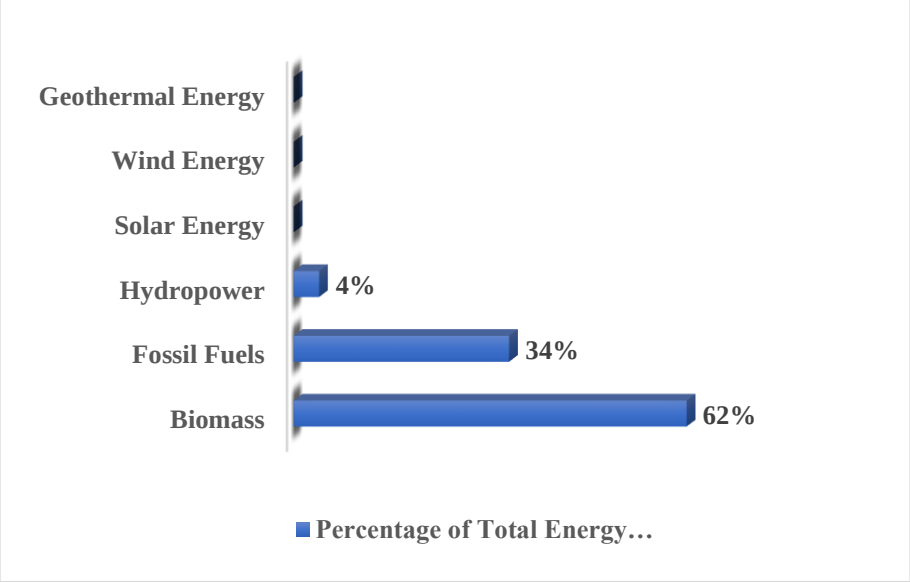

Figure 2.2: Sources of Energy Supply in Sudan

Renewable energy sources (solar, wind, and geothermal) make up a very small percentage but hold substantial untapped potential.

Electricity access in Sudan is highly uneven (illustrated in Figure 2.3), with urban areas significantly better served than rural regions, where many households lack grid connections. The national grid, consisting of the Blue Nile and Western grids, covers only a small portion of the population, leaving vast areas dependent on off-grid solutions like diesel generators. The situation is worsened by frequent power outages, outdated infrastructure, and insufficient investment in grid maintenance.(Fadlallah & Benhadji Serradj, 2020)

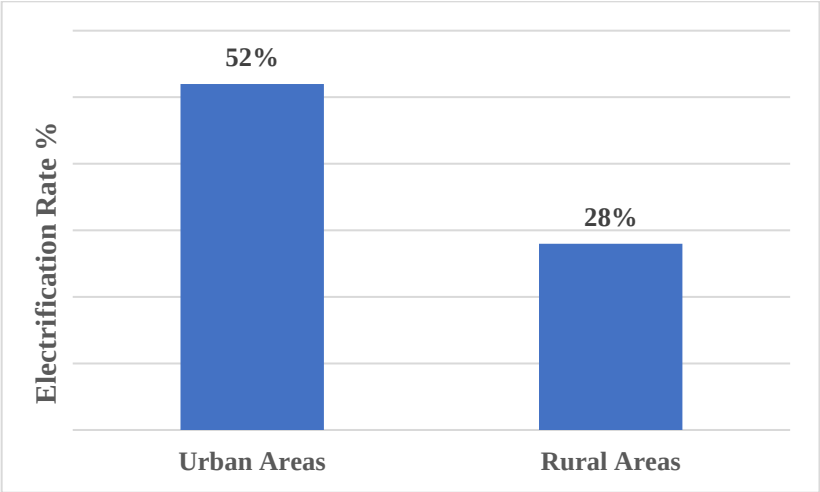

Figure 2.3: Electrification Rate in Sudan

Sudan's grid infrastructure is further hampered by political instability and economic challenges. High inflation, currency devaluation, and fluctuating global oil prices have hindered the government's ability to fund large-scale energy projects. The ongoing conflict has disrupted fuel supply chains, creating persistent shortages that severely impact energy-dependent sectors such as agriculture and transportation.(N. Ibrahim, 2024)

2.3.2 Opportunities for Solar Energy Integration

Sudan has a very high potential for solar energy, (as can be seen from Figure 2.4). The solar generation capacity rises from the south to the north. The northern regions tend to have higher solar irradiance whereas higher sunlight is also linked with the arid climate. Therefore, despite having a huge area available for crop cultivation, the arid nature of the terrain limits the potential farming. At the same time, there are huge reserves of underground water which can potentially help to convert the arid land into fertile pastures.

Conventional rain-fed irrigation can be converted into controlled irrigation by providing the farmers with opportunities to use underground water when and how they need it. That's why solar water pumping offers a promising solution for addressing the country's irrigation challenges. By replacing diesel pumps, solar-powered systems provide a consistent water supply, lower operational costs, and significantly reduce greenhouse gas emissions. Studies have shown that integrating solar-powered irrigation can increase agricultural productivity by up to 25%, particularly in areas where water shortages constrain crop yields.(B. Ali, 2018)

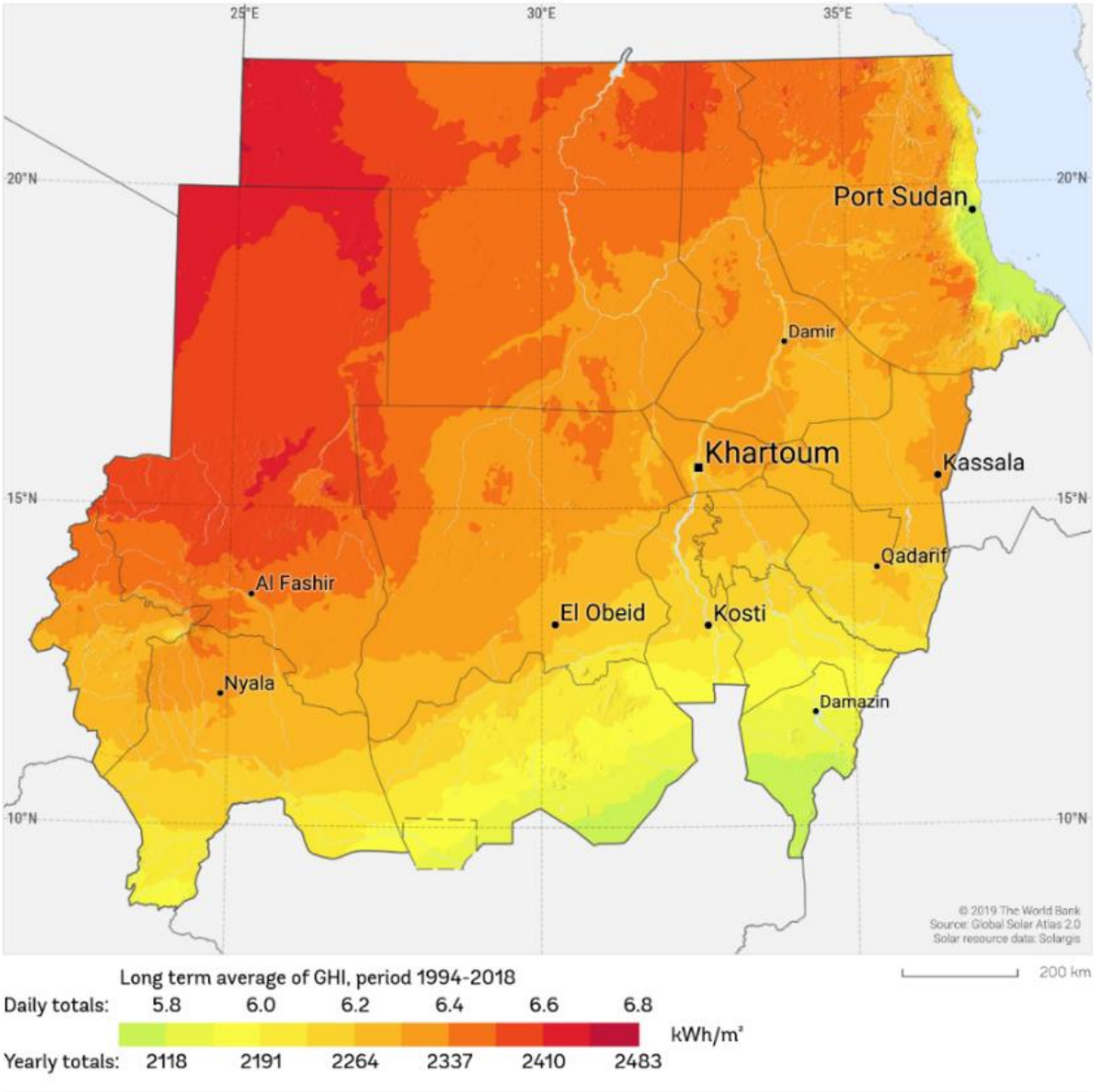

Figure 2.4: Solar capacity map of Sudan

In addition, solar water pumping aligns with Sudan's long-term sustainability goals. Its decentralized nature allows for deployment in remote regions, where grid access is limited. Farmers in such areas could benefit from stable irrigation cycles, improved crop reliability, and reduced dependence on fuel supply chains that are vulnerable to price fluctuations and logistical delays.(B. Ali, 2018)

2.4 Comparative Analysis: Diesel vs. Solar Water Pumping

In Sudan's agricultural sector, both diesel and solar water pumping systems can be used to address irrigation needs, each offering distinct advantages and drawbacks. While diesel pumps remain prevalent due to their lower upfront costs and familiarity, solar water pumps are increasingly recognized for their long-term economic and environmental benefits. This section examines the two systems across three key factors: economic considerations, environmental impact, and reliability, highlighting their suitability for Sudan's agricultural landscape.(B. Ali, 2018)

2.4.1 Economic Considerations

Diesel-powered pumps are often selected for their relatively low initial costs, with a typical system priced between $500 and $15,000, depending on capacity. This affordability makes them accessible to smallholder farmers in Sudan, who often face financial constraints and lack access to credit. However, the operational costs of diesel systems are high. Fuel expenses alone can account for up to 40% of a farmer's annual irrigation budget, with diesel prices in Sudan fluctuating due to political instability and supply chain issues. For instance, the country's frequent fuel shortages and high inflation have caused diesel prices to spike, further burdening farmers.(B. Ali, 2018)

In contrast, solar water pumping systems require a higher upfront investment, ranging from $2,000 to more than $30,000 depending on the scale and location of installation. Despite the higher initial cost, solar systems offer negligible operational expenses as they rely on freely available sunlight. A study by the International Renewable Energy Agency (IRENA) indicates that solar systems in Sudan can achieve cost recovery within 3–5 years, saving farmers between $500 and $1,200 annually compared to diesel systems. Over a 10-year lifecycle, solar systems consistently prove more economical, particularly in regions with abundant sunlight, which Sudan enjoys year-round.(B. Ali, 2018)

2.4.2 Environmental Impact

Each liter of diesel burned emits approximately 2.68 kilograms of CO_2. For irrigation, the CO_2 emissions per cubic meter of water pumped typically range from 0.5 to 1.5 kilograms of CO_2, depending on system efficiency and pumping conditions. Additionally, diesel usage contributes to air and soil pollution, with spills during transport and storage exacerbating environmental degradation in rural farming communities.(El-Gafy & El-Bably, 2016)

Solar water pumps, in contrast, operate with zero direct emissions, aligning with global climate goals. By replacing diesel pumps, a solar system in Sudan can prevent the release of up to 1.5 tons of CO_2 per year, depending on usage. Solar water pumping systems also contribute to sustainability by reducing dependency on imported diesel, which is vulnerable to price volatility and supply disruptions. Their adoption aligns with Sudan's commitment to reducing emissions as part of broader international climate agreements.(Nasir, 2019)

2.4.3 Reliability and Maintenance

Diesel-powered systems are often praised for their reliability, especially in scenarios where sunlight availability may be intermittent. However, maintaining diesel pumps presents significant challenges for farmers in Sudan. Regular maintenance, such as oil changes, filter replacements, and repairs, adds to operational costs and often results in prolonged downtime. In remote regions, where access to spare parts and technical expertise is limited, these challenges are compounded.(Aliyu et al., 2018)

Solar water pumping systems, while dependent on sunlight, offer greater reliability over their lifespan. Modern systems often include battery storage to ensure functionality during cloudy periods or at night, reducing reliance on diesel backups. Maintenance requirements for solar systems are minimal, involving periodic cleaning of panels and inspections of electrical connections. These features make solar systems particularly well-suited for Sudan's rural and off-grid areas, where technical resources are scarce and maintenance costs must be minimized. (B. Ali, 2018)

The comparative advantages of solar water pumping, particularly its economic and environmental benefits, underscore its potential as a transformative technology for Sudan's agricultural sector. However, understanding the broader implications of these systems requires a closer examination of how they interact with Sudan's unique energy policies,

agricultural practices, and socio-economic factors. The following section delves into these dynamics, exploring the integration of renewable energy solutions within the Sudanese context to address persistent irrigation challenges.(B. Ali, 2018)

2.5 Socioeconomic Impacts of Solar Water Pumping

The ongoing conflict in Sudan has exacerbated the challenges faced by smallholder farmers, including severe diesel shortages, rising fuel prices, and disruptions to food production and supply chains. In this context, solar water pumping systems offer a transformative solution, providing farmers with a sustainable means of irrigation while reducing dependency on scarce and expensive fossil fuels. This section explores the role of solar water pumping in addressing these hardships, focusing on the opportunities for enhancing agricultural resilience and supporting rural livelihoods amidst the ongoing crisis.(B. Ali, 2018)

Addressing Diesel and Food Shortages

Diesel shortages in Sudan have become a critical issue for farmers who rely on diesel-powered pumps for irrigation. The conflict has disrupted fuel supply chains, leading to significant price increases and frequent unavailability of fuel in rural areas. Solar water pumping systems provide a reliable alternative by leveraging the country's abundant solar resources. These systems enable farmers to maintain irrigation cycles without the uncertainty of fuel availability, ensuring consistent crop production during periods of crisis.(Khan, Imran, Umer, et al., 2021)

By stabilizing water access, solar water pumps contribute to mitigating food shortages that have been intensified by the conflict. Farmers equipped with solar systems are better positioned to grow staple crops, supporting local food security and reducing reliance on imports, which are often disrupted by border closures and logistical challenges. This stability in food production can alleviate the strain on rural communities, which are among the hardest hit by the conflict's economic and social impacts.(Falchetta et al., 2023)

Solar water pumping systems also enhance agricultural resilience by reducing operational costs associated with diesel-powered systems. The elimination of recurring fuel expenses allows farmers to redirect limited financial resources toward purchasing seeds, fertilizers, and other inputs necessary for maintaining or increasing productivity. Additionally, solar-powered irrigation reduces the dependency on rain-fed agriculture, enabling farmers to

adapt to the erratic rainfall patterns that have worsened due to climate change.(Khan, Imran, Umer, et al., 2021)

In conflict-affected areas where access to maintenance services for diesel pumps is limited, solar systems offer a significant advantage. Their low maintenance requirements and durability make them particularly suited for remote and resource-scarce environments. This reliability ensures farmers can continue their operations with minimal interruptions, even in challenging circumstances.(Khan, Imran, Umer, et al., 2021)

Barriers to Adoption

Despite these advantages, the adoption of solar water pumping systems is hindered by several barriers, which are amplified by the ongoing conflict. Financing remains a critical issue, as many farmers lack the capital to invest in solar systems or the ability to access credit due to the weakened banking system. The conflict has also disrupted government programs and international aid initiatives that could support renewable energy adoption, further limiting farmers' options.

Technical expertise and capacity are additional challenges. The conflict has displaced skilled technicians and disrupted supply chains for solar equipment, making it difficult for farmers to install and maintain solar systems. Awareness campaigns and training programs, already limited before the conflict, have been further hampered, leaving many farmers unaware of the potential benefits of transitioning to solar-powered irrigation.

Policy and Institutional Support for Solar Water Pumping

Solar water pumping systems hold the potential to provide stability amidst Sudan's ongoing conflict by reducing reliance on diesel and supporting local food production. However, realizing this potential requires robust policy and institutional support to address the systemic barriers to adoption. Financing mechanisms, technical capacity building, and prioritization of renewable energy solutions in agricultural policies are essential for scaling up solar water pumping and ensuring its long-term impact.(B. Ali, 2018)

Global experiences highlight the importance of well-designed policies in accelerating the adoption of solar water pumping systems. Subsidies, tax incentives, and accessible financing programs have been instrumental in enabling farmers to transition from diesel to solar-powered irrigation in regions like India and East Africa. In contrast, Sudan's policy landscape remains fragmented, with limited provisions for promoting renewable energy in agriculture. The lack of cohesive strategies leaves farmers without the necessary

incentives or support to invest in solar systems, increasing reliance on fossil fuels.(Abdalla, 2021; Falchetta et al., 2023)

Institutional barriers further hinder progress. The absence of structured training programs and technical support networks prevents farmers from understanding and maintaining solar water pumping systems. Additionally, the limited engagement of local governments and stakeholders in renewable energy initiatives creates gaps in awareness and resource allocation. Strengthening institutional frameworks and fostering partnerships with international organizations could help bridge these gaps, making solar technology more accessible and effective in addressing Sudan’s agricultural challenges.(Abdalla, 2021; Khan, Imran, Umer, et al., 2021)

2.6 Conclusion

The review of existing literature reveals that solar-powered irrigation systems hold substantial promise for enhancing agricultural productivity, improving rural energy access, and advancing climate resilience—particularly in arid, off-grid regions like the River Nile State in Sudan. Globally, solar water pumping has demonstrated technical viability and economic benefits, especially in contexts where diesel-based systems are proving increasingly unsustainable. Regional experiences across sub-Saharan Africa underscore the importance of aligning technology deployment with financing, training, and policy frameworks to ensure long-term impact.

Despite Sudan's high solar potential and urgent agricultural needs, adoption of solar irrigation remains limited. Smallholder farmers face multiple barriers, including high upfront costs, lack of access to credit, limited technical know-how, and weak institutional support. Moreover, existing studies often lack localized, data-driven analyses that account for Sudan's specific environmental, social, and economic conditions. While international models and pilot projects offer valuable lessons, they cannot fully substitute for context-specific evaluations grounded in local realities.

This gap forms the foundation for the present research. By using CROPWAT and HOMER Pro to assess the technical and economic feasibility of transitioning from diesel to solar irrigation for a small-scale farm in the River Nile State, the study aims to provide evidence-based insights that can inform both policy and practice. It seeks to demonstrate under what conditions solar water pumping becomes a viable and sustainable alternative for Sudanese farmers, and what kinds of support mechanisms: financial, technical, or institutional, are necessary to enable that transition. In doing so, the research contributes not only to the academic discourse on renewable energy in agriculture but also to the practical advancement of sustainable rural development in Sudan.

Chapter Three

Methodology

3 Methodology

3.1 Study Approach and Methodological Framework

The methodology adopted in this thesis integrates both quantitative and analytical approaches to evaluate the technical and economic feasibility of transitioning from diesel-powered to solar-powered irrigation systems in Abu Hamed, River Nile State, Sudan. Given the limited availability of field data and real-time observations due to regional constraints, the study predominantly employs secondary data and theoretical estimations, supported by specialized computational software.

The methodological design consists of clearly defined stages:

- **Data Acquisition and Preparation:** Utilizing comprehensive climatic, agricultural, and economic data sources tailored to the conditions of Abu Hamed.
- **Crop Water Requirement Estimation:** Quantifying crop evapotranspiration, irrigation schedules, and net irrigation requirements by integrating climatic, crop-specific, and soil data.
- **Energy Demand Calculation:** Estimating the energy required for irrigation based on calculated irrigation needs, pump efficiency, and operating conditions.
- **Energy System Simulation and Comparative Analysis:** Evaluating the performance, economic feasibility, and environmental impacts of solar versus diesel-powered irrigation systems.

To implement these methodological steps effectively, the study employs specialized computational software. The chosen software packages (CROPWAT, CLIMWAT, and HOMER Pro) provide robust analytical capabilities, accurate modeling, and optimization features essential for achieving the study's objectives. Their specific functionalities, reasons for selection, and relevance to the methodological framework are detailed in the following subsections.

3.1.1 CROPWAT and CLIMWAT

CROPWAT and CLIMWAT are software tools developed by the Food and Agriculture Organization (FAO) of the United Nations, widely used for irrigation planning and agricultural water management. They enable users to calculate precise irrigation requirements based on climate, crop, and soil data, thus optimizing water use efficiency.(Shamsuddin & Mohamed Amran, 2016)

CROPWAT

It specifically calculates crop water requirements, irrigation schedules, and water supply schemes based on the Penman-Monteith equation. This capability allows for accurate quantification of the irrigation water needed for specific crops throughout their growing cycles. CROPWAT's outputs, such as crop evapotranspiration ETc and net irrigation water requirements $NIWR$, form critical inputs for energy demand estimation.

CLIMWAT

It complements CROPWAT by providing region-specific climatic data, including temperature, humidity, wind speed, solar radiation, and rainfall averages from meteorological stations worldwide. CLIMWAT ensures that climatic inputs into the model are precise and relevant to Abu Hamed, enhancing the accuracy and reliability of water requirement estimations.

The integration of CROPWAT and CLIMWAT was essential due to their proven effectiveness in agricultural planning, their ability to handle limited or region-specific data, and their widespread acceptance in international water management studies.

Equations Used by software:

Reference Evapotranspiration (ET_0):

CROPWAT calculates ET_0 using the Penman-Monteith equation **3.1**:

$$\boldsymbol{ET_0 = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T + 273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)}} \qquad 3.1$$

Where:

- $\boldsymbol{\Delta}$ = Slope of the saturation vapor pressure curve (kPa/°C)
- $\boldsymbol{R_n}$= Net radiation at the crop surface (MJ/m²/day)
- $\boldsymbol{G}$ = Soil heat flux density (MJ/m²/day)
- $\boldsymbol{\gamma}$ = Psychrometric constant (kPa/°C)
- $\boldsymbol{T}$ = Mean daily air temperature (°C)
- $\boldsymbol{u_2}$= Wind speed at 2 meters (m/s)
- $\boldsymbol{e_s - e_a}$= Vapor pressure deficit (kPa)

3.1.2 HOMER Pro

HOMER Pro, developed by the National Renewable Energy Laboratory (NREL), is a comprehensive simulation and optimization software used to design, analyze, and compare renewable and conventional energy systems. Its primary strength lies in its ability to evaluate the technical and economic viability of various energy configurations, making it particularly suitable for assessing the feasibility of renewable energy solutions in remote and off-grid settings.

In this study, HOMER Pro was utilized to simulate the operational and financial performance of two distinct irrigation energy systems: solar photovoltaic (PV) and diesel-powered pumping. The software enabled precise calculation of critical economic indicators, such as Net Present Cost NPC and Levelized Cost of Energy $LCOE$, as well as the system's environmental impact, particularly carbon emissions associated with diesel-powered irrigation.

Furthermore, HOMER Pro facilitated the simulation of system performance under various scenarios, allowing detailed analysis of the solar PV system's energy output, reliability, and financial benefits relative to the diesel alternative. This analytical capability was critical for providing clear and scientifically rigorous insights into the economic and environmental advantages of transitioning to solar energy in Abu Hamed.

Equations Used by software:

1. **Levelized Cost of Energy $LCOE$:**

HOMER Pro calculates $LCOE$, which represents the cost per kWh of energy:

$$\boldsymbol{LCOE} = \frac{\boldsymbol{Total\ Net\ Present\ Cost\ (NPC)}}{\boldsymbol{Total\ Energy\ Production}} \qquad 3.2$$

2. **Carbon Emissions:**

Diesel system emissions are estimated based on fuel consumption:

$$\boldsymbol{CO_2} = \boldsymbol{Fuel\ Consumption\ (liters)} \times \boldsymbol{Emission\ Factor\ (kg\ CO_2/liter)} \qquad 3.3$$

The study design is grounded in a quantitative, computational approach, allowing for precise assessment of system performance under varying scenarios. This approach ensures that the analysis remains scientifically rigorous while addressing practical challenges specific to the River Nile State.

Additionally, the phased design facilitates a step-by-step evaluation of both technologies, enabling the identification of key factors that influence their feasibility.

The following sections further elaborate on the specific steps involved in data collection, preparation, and analytical modeling, building upon the robust computational foundation established by these software tools.

3.2 Data Collection and Preparation

The data collection process is central to the study, ensuring the reliability and validity of calculations for crop water requirements, energy demands, and system comparisons. This section outlines the step-by-step approach to gathering and preparing the climatic, crop, soil, and energy data required for analysis. Assumptions are also explicitly detailed to clarify the basis for calculations and address potential limitations.

3.2.1 Sources of Data

3.2.1.1 Climatic Data

Climatic data is fundamental to estimating crop water requirements, as it directly influences evapotranspiration rates. Evapotranspiration ET is the combined process of water loss from the soil surface through evaporation and from plant surfaces through transpiration. This parameter is critical for determining the amount of water a crop needs to grow optimally.(Al Zayed et al., 2016)

For this study, climatic parameters were obtained from the following sources:

CLIMWAT Database

It provides region-specific meteorological data, including temperature, solar radiation, wind speed, and humidity for the River Nile State.

Supplementary Sources

For parameters unavailable or outdated in CLIMWAT, data was referenced from global meteorological databases, such as the FAO Climate Data repository.

The selected parameters include:

Temperature

Used to calculate daily evapotranspiration ET_0 as it affects crop water loss rates.

Humidity

Influences the water vapor deficit and ET_0 values.

Wind Speed

Affects the evaporation rate of water from soil and crop surfaces.

Solar Radiation

Determines the amount of energy available for crop growth and is essential for solar pump energy modeling.

Rainfall

Used to calculate effective rainfall, defined as the portion of total rainfall that is available for crop use after accounting for losses such as surface runoff.(Aschale et al., 2023)

3.2.1.2 Crop Data

Crop-specific information was collected to model the water requirements of wheat, the selected crop for this study.

The data includes:

Planting and Growth Stages

Determined from FAO guidelines and regional agricultural calendars to align with typical cropping cycles in the River Nile State.

Crop Coefficient (Kc)

This dimensionless factor adjusts the reference evapotranspiration ET_0 to account for the specific water needs of a crop at different growth stages. For example, wheat has distinct water requirements during its germination, vegetative growth, and flowering stages.

Rooting Depth

Refers to the depth within the soil from which the crop extracts water. This parameter affects the volume of soil moisture accessible to the plant and, consequently, irrigation requirements.(Yang et al., 2024)

3.2.1.3 Soil Data

Soil properties play a critical role in determining irrigation efficiency and water retention capacity.

The following parameters were collected:

Soil Texture

Refers to the composition of soil based on the proportions of sand, silt, and clay, which affects infiltration and water-holding capacity.

Available Water Content (AWC)

This is the amount of water that the soil can store and release for crop use. It is calculated as the difference between the soil's field capacity and its wilting point.

Infiltration Rate

Indicates how quickly water can penetrate the soil surface, influencing irrigation efficiency.

Field Capacity

The maximum amount of water the soil can hold after excess water has drained.

Wilting Point

The soil moisture level below which crops cannot extract water, leading to water stress.

Soil data was sourced from regional profiles in FAO databases and verified using guidelines from the USDA soil conservation service.(Adam et al., 2021)

3.2.1.4 Energy Data

Energy data is required to compare diesel and solar pumping systems.

The following information was collected:

Fuel Consumption and Prices

Data on diesel pump efficiency and fuel costs were obtained from regional energy market reports and agricultural records.

Solar Irradiation

Solar potential maps of Sudan, including specific data for the River Nile State, were used to model the feasibility of solar systems. Solar irradiation refers to the total amount of

solar energy received per unit area, typically expressed in kilowatt-hours per square meter per day ($kWh/m^2/day$).

Pumping Efficiency

Refers to the percentage of input energy that is effectively converted into useful work (in this case, water pumping). This depends on the type and condition of the pump used. (B. Ali, 2018)

3.2.2 Data Preparation

3.2.2.1 Climatic Data Preparation

Formatting for CROPWAT

Meteorological parameters such as temperature, humidity, wind speed, and solar radiation were formatted into monthly averages for compatibility with the CROPWAT software.(Surfia Dioh, 2022)

ET_0 Calculation

ET_0 values were calculated using the Penman-Monteith equation, which integrates multiple climatic parameters to determine baseline water loss rates. The Penman-Monteith equation is widely regarded as the standard method for calculating reference evapotranspiration in agricultural studies.(Djaman et al., 2019)

3.2.2.2 Crop Parameter Integration

Growth Stage Alignment

The planting dates and growth stages of wheat were matched with regional agricultural calendars to ensure realistic simulation.(SHARMA et al., 2024)

Crop Coefficients (Kc)

Stage-specific Kc values were applied to adjust ET_0 to ETc, accounting for changes in crop water use during germination, vegetative growth, flowering, and maturity.(Yang et al., 2024)

3.2.2.3 Soil Profile Adjustment

Net Irrigation Water Requirement (NIWR)

Soil moisture characteristics (field capacity and wilting point) were used to estimate the net irrigation water requirements $NIWR$. $NIWR$ is defined as the total amount of water required to meet crop water needs, adjusted for effective rainfall and soil properties.(Kukal & Irmak, 2023)

Effective Rainfall Calculation

Effective rainfall, the portion of rainfall available for crop use, was subtracted from ETc to determine irrigation needs. This calculation assumed minimal surface runoff and even rainfall distribution.(M. Ali & Mubarak, 2017)

3.2.2.4 Energy System Inputs

Energy Demand Estimation

Water flow rates and pump head (including static lift and friction losses) were calculated to determine the energy required for irrigation. Pump head refers to the total height that the water must be lifted, including vertical lift and frictional losses in the pipes.(San et al., 2019)

System Modeling

Solar irradiation data was integrated with HOMER Pro to simulate solar pumping system performance, while diesel fuel consumption rates were used to model operational costs and emissions of diesel systems.(Elkadeem et al., 2019)

3.2.3 Assumptions

1. **Climatic Uniformity**

Climatic conditions in the River Nile State were assumed to remain stable throughout the study period, with monthly averages representing typical weather patterns.

2. **Uniform Soil Characteristics**

Soil properties were assumed to be homogeneous across the study area due to a lack of granular field data.

3. **Irrigation Efficiency**

A standard irrigation efficiency was assumed for both diesel and solar systems to account for water losses during distribution and application.

4. **Diesel Price Stability**

Diesel prices were assumed to remain constant during the study period, acknowledging that price fluctuations could affect the cost-effectiveness of diesel systems.

5. **Optimal Pump Efficiency**

Pumps were assumed to operate at their rated efficiency levels, with no significant degradation over the study period.

3.3 Analytical Framework

The framework consists of three core components: estimating crop water requirements, calculating energy demand, and modeling system performance. These components work together to provide a holistic evaluation of the irrigation solutions under study.

3.3.1 Estimation of Crop Water Requirements

Determining how much water crops need is fundamental to designing efficient irrigation systems. To achieve this, the study uses CROPWAT 8.0, a software developed by the Food and Agriculture Organization (FAO). CROPWAT calculates crop water requirements by integrating climatic, crop, and soil data, allowing for accurate irrigation planning.(Khan, Imran, Umer, et al., 2021)

Key Inputs and Calculations:

Climatic Data

Parameters such as temperature, humidity, wind speed, solar radiation, and rainfall are collected from the CLIMWAT database and regional meteorological records. These factors directly impact the Reference Evapotranspiration ET_0, which measures the rate at which water evaporates and transpires from crops under ideal conditions.(Mattar & Alazba, 2019)

Crop Data

Information on planting dates, crop growth stages, and Crop Coefficients Kc is used to adjust ET_0 for wheat cultivation, resulting in the Crop Evapotranspiration ETc.(Yang et al., 2024)

Soil Data

Soil texture, infiltration rate, available water content, field capacity, and wilting point are included to determine how much water the soil can retain and supply to crops.(Goh Eng Giap et al., 2021)

The crop water requirement is calculated using equation **3.4**:

$$ET_c = K_c \times ET_0 \qquad 3.4$$

To avoid overestimating irrigation needs, Effective Rainfall R_{eff} is subtracted from ETc to calculate the Net Irrigation Water Requirement $NIWR$, equation 3.5**3.5**:

$$\boldsymbol{NIWR = ET_c - R_{eff}} \qquad 3.5$$

3.3.2 Energy Demand Estimation

The energy required for irrigation was estimated by converting the net irrigation water requirement $NIWR$ into the required pumping power, considering the total dynamic head TDH, system efficiency, and operational duration. This approach ensures that the system is correctly sized to meet peak irrigation demand while optimizing energy consumption.

Pump Flow Rate Calculation

The required pump flow rate was derived based on the peak irrigation demand during the highest water requirement month. The daily irrigation volume $\boldsymbol{Q}$**.**

was calculated using:

$$\boldsymbol{Q = \frac{NIWR \times A}{1000}} \qquad 3.6$$

Where:

- Q = Daily irrigation water volume (m^3/s)
- $NIWR$ = Peak net irrigation requirement (l/s/h)
- A = Irrigated field area (hectares)
- 1000 = Conversion factor from liter to m^3

The **total dynamic head H_{TDH}** was estimated based on:

$$H_{\mathrm{TDH}} = H_s + H_f \qquad 3.7$$

Where:

- H_s **(Static Lift):** Depth from which water is drawn
- H_f **(Friction Losses):** Estimated based on pipe length, diameter, and flow rate

Pump Power Calculation

The required **pump power (P)** was calculated using the hydraulic power equation:

$$P = \frac{Q \times H_{\mathrm{TDH}} \times g \times \rho}{\eta \times 3600} \qquad 3.8$$

Where:

- P = Pump power (kW)
- Q = Pump flow rate (m^3/h)
- H_{TDH} = Total dynamic head (m)
- g = Gravitational acceleration (9.81 m/s^2)
- ρ = Water density (1000 kg/m^3)
- η = Pump efficiency
- 3600 = Conversion factor for time units

The calculated peak power demand P was then used as the baseline energy consumption input for HOMER simulations.

3.3.3 System Modeling and Simulation

To analyze and compare the performance of diesel and solar-powered pumping systems, the study utilizes HOMER Pro software. It models system configurations, evaluates their feasibility, and conducts financial and environmental analyses.(Elkadeem et al., 2019)

Modeling Steps:

1. Energy Demand Input

The calculated energy requirement is entered into HOMER Pro to simulate system operations.

2. System Configuration

Two system designs are modeled—one using solar photovoltaic (PV) panels with storage and the other using diesel generators.

3. Economic Analysis

HOMER Pro evaluates system costs over the project's lifespan by calculating the Net Present Cost NPC and the Levelized Cost of Energy $LCOE$.

4. Environmental Impact Assessment

From the software results carbon emissions for the diesel system can be estimated, showing the direct environmental footprint of diesel system.

3.3.4 Integration of Data and Process Flow

To illustrate the methodological flow, the following diagrams **Figure 3.1** and **Figure 3.2** summarizes the step-by-step process from data collection to system modeling:

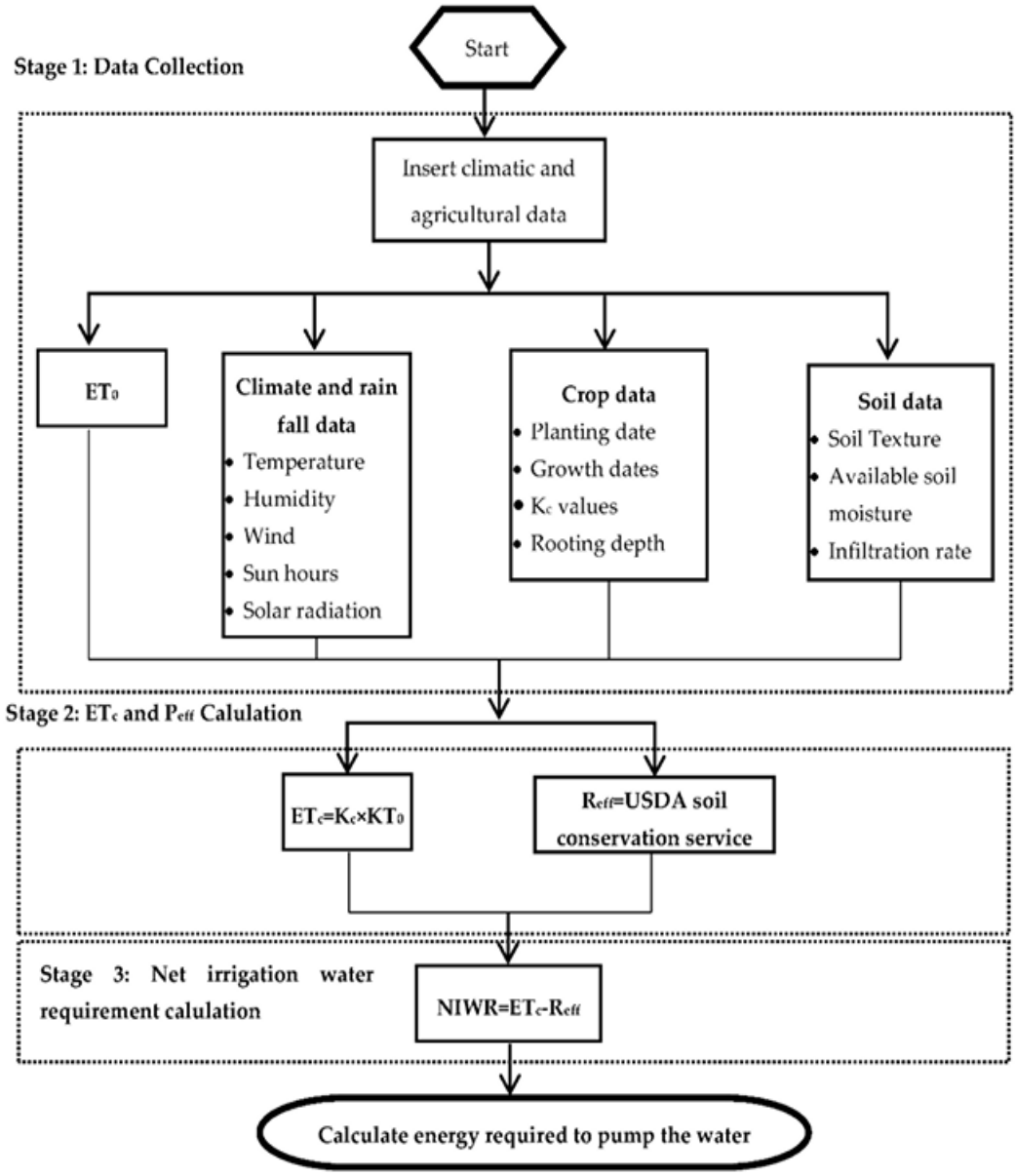

Figure 3.1: Flow chart for the crop water requirement and energy calculation.

(Khan, Imran, Umer, et al., 2021)

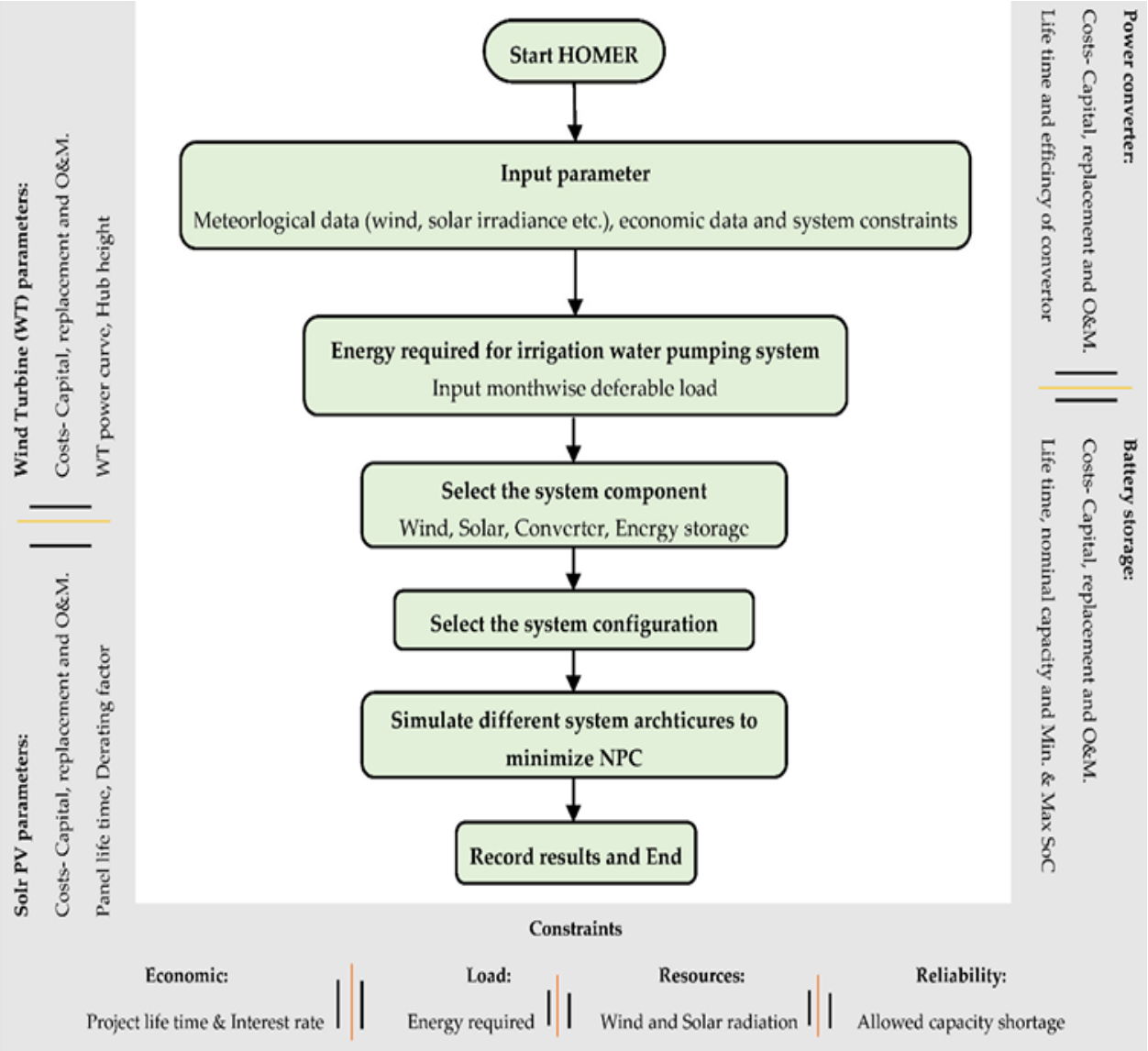

Figure 3.2: Homer process flow for the optimization of the renewable system.

(Khan, Imran, Altamimi, et al., 2021)

3.4 Limitations

While this study provides valuable insights into the feasibility of transitioning from diesel-powered to solar-powered irrigation systems in Sudan's River Nile State, several limitations must be acknowledged to provide a balanced perspective on the research findings:

Data Constraints

The study heavily relies on secondary data sources due to the lack of recent and localized field data. This reliance on publicly available datasets may limit the accuracy and representativeness of the results, particularly in capturing local climatic variations and soil characteristics.

Simplified Modeling

The computational models used, including CROPWAT and HOMER Pro, involve theoretical calculations that may not fully capture real-world complexities. Factors such as system inefficiencies, equipment wear and tear, and unpredictable operational challenges are not fully represented in the models.

Outdated Data

Some of the climatic, soil, and energy market data utilized in the analysis may be outdated. Changes in weather patterns, soil degradation, and fluctuations in fuel prices could influence the accuracy of the water requirement estimations and system performance simulations.

Unverified Field Conditions

Due to limited access to on-ground measurements, several assumptions were made regarding soil uniformity, irrigation efficiency, and pump performance. These assumptions may not accurately reflect the actual conditions in the River Nile State, potentially impacting the reliability of the findings.

Economic Volatility

The study assumes stable diesel fuel prices throughout the analysis period. However, fuel markets are subject to significant fluctuations, which could impact the economic viability of diesel-powered systems and alter the comparative cost-effectiveness between diesel and solar options.

Technological Barriers

The research does not account for potential operational and maintenance challenges associated with solar-powered pumping systems. In remote areas, limited access to technical support and spare parts could hinder the effective deployment and long-term operation of solar systems.

Chapter Four

Case Study

4 Case Study

4.1 Site Selection

The selection of Abu Hamed in the River Nile State, Sudan, as the site for implementing solar-powered water pumping systems for irrigation is a strategic decision supported by its geographical advantages, favorable climatic conditions, and substantial agricultural potential. The case study focuses on a 10-hectare farm in Abu Hamed, emphasizing its suitability for adopting renewable energy solutions in agriculture. This specific location offers a valuable opportunity to evaluate the effectiveness and economic feasibility of transitioning from diesel-powered to solar-powered irrigation systems, considering the site's climatic and resource conditions. The study aims to provide insights into how solar energy can enhance agricultural productivity while reducing operational costs and environmental impact.

4.1.1 Geographical Location

Abu Hamed is strategically positioned in northern Sudan along the eastern bank of the Nile River, approximately **555** kilometers north of Khartoum. Its precise geographical coordinates are **19°32′36″N** latitude and **33°20′16″E** longitude, placing it within Sudan's arid desert zone. The town's proximity to the Nile River is a critical advantage, providing a continuous and sustainable water source essential for irrigation in this otherwise water-scarce environment. Additionally, the flat terrain in Abu Hamed is favorable for the installation of solar photovoltaic (PV) systems and irrigation infrastructure, reducing construction complexity and costs.(Hassabelgabo Abdelrazig Ibrahim & Mohammed Elmardi Suleiman Khayal, 2020)

4.1.2 Climatic Conditions

All climatic data for this study were collected from CLIMWAT, a reliable source for climate data used in agricultural planning. The collected data were utilized to calculate the reference evapotranspiration (ET_0) for each month using the FAO Penman-Monteith equation embedded in CROPWAT software, utilizing equation 3.1. This ET_0 calculation provides a precise estimate of the water demand for crops, forming the basis for irrigation planning in this region.

Month	Min Temp	Max Temp	Humidity	Wind	Sun	Rad	ETo
	°C	°C	%	km/day	hours	MJ/m²/day	mm/day
Jan	12.5	28.1	33	501	8.9	17.9	7.37
Feb	13.8	30.4	25	501	9.3	20.3	8.75
Mar	17.3	34.6	20	467	9.8	23.1	10.27
Apr	21	38.7	18	423	10.5	25.4	11.33
May	25	41.9	17	346	10.5	25.7	11.17
Jun	27.2	43.5	16	346	10.7	25.9	11.7
Jul	26.9	42.1	23	346	9.9	24.6	10.97
Aug	27.2	41.7	30	346	9.5	23.9	10.34
Sep	27.3	42.4	21	423	9.3	22.7	11.98
Oct	23.7	39.2	24	423	9.8	21.4	10.68
Nov	18.9	33.3	33	501	9.2	18.6	8.94
Dec	14.4	29.4	36	501	8.9	17.2	7.45
Average	21.3	37.1	25	427	9.7	22.2	10.08

Table 4.1: Monthly ET_0 Penman-Monteith, Abu Hamed Station

Abu Hamed's climate is classified as hot desert (BWh) under the Köppen climate classification. This region experiences extreme temperatures, particularly during the summer months. In June, temperatures peak at **43.5°C**, while winter months are milder, with average highs of **28.1°C** in January. These high temperatures contribute to significant evapotranspiration (ET_0), which increases the water demand for crops and underscores the need for efficient irrigation systems. Additionally, Abu Hamed enjoys **9 to 11 hours** of sunlight daily, ensuring consistent and reliable performance of solar energy systems throughout the year.(Khan, Imran, Umer, et al., 2021)

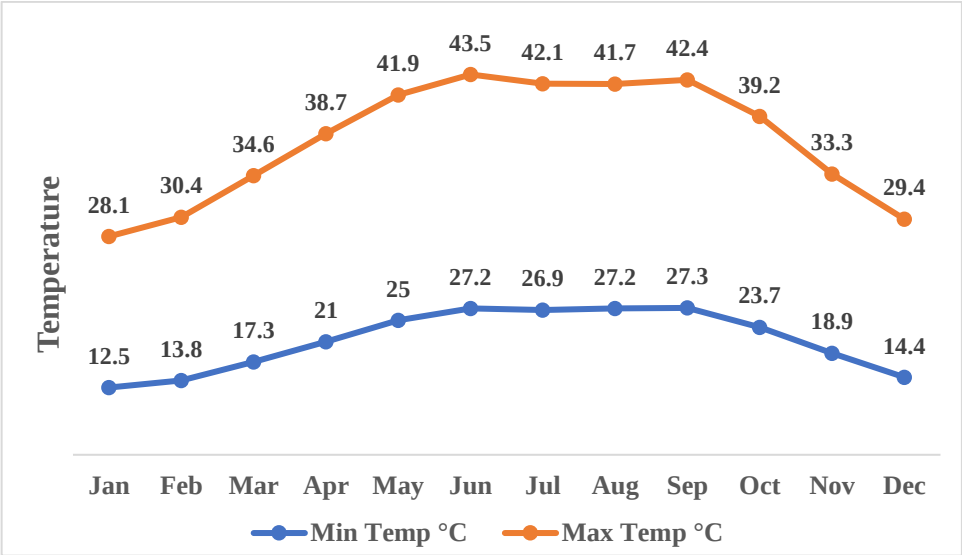

Figure 4.1: Average Temperature in Abu Hamed, Sudan

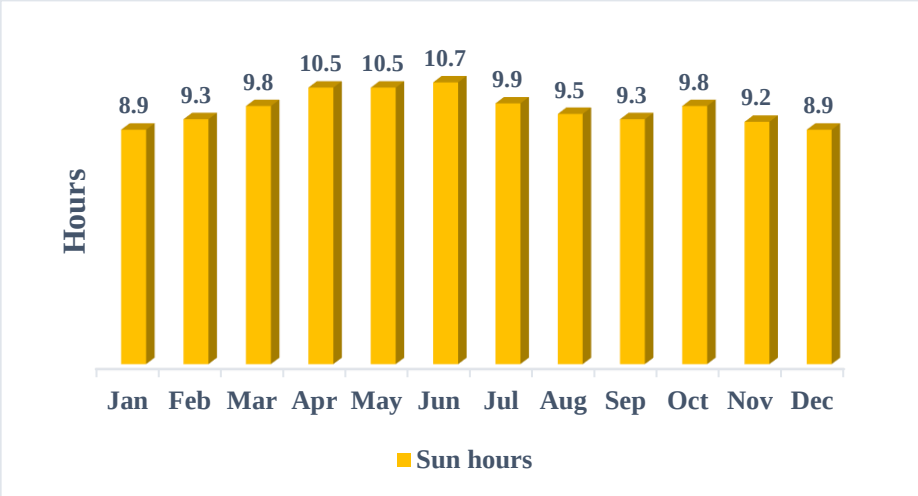

Figure 4.2: Sunshine Duration in Abu Hamed

Building on the analysis of temperature and sun hours in Abu Hamed (illustrated in Figure 4.1 and Figure 4.2), the region's exceptional solar energy potential further enhances its suitability for renewable energy applications. Solar radiation levels, a critical factor for solar-powered systems, are particularly remarkable in Abu Hamed. The measured values range from **17.2 MJ/m²/day** in December to a peak of **25.9 MJ/m²/day** in June, with an annual average of **22.2 MJ/m²/day**. These high solar radiation levels align seamlessly with the long daily sun hours discussed earlier, ensuring optimal energy generation throughout the year.

The seasonal variation in solar radiation is relatively moderate (see Figure 4.3), with consistently high levels observed during the summer months (April to July) when agricultural water demands are at their peak. Even in the winter months (November to February), when solar radiation dips slightly, it remains sufficient to support the energy needs of irrigation systems.(Hassabelgabo Abdelrazig Ibrahim & Mohammed Elmardi Suleiman Khayal, 2020)

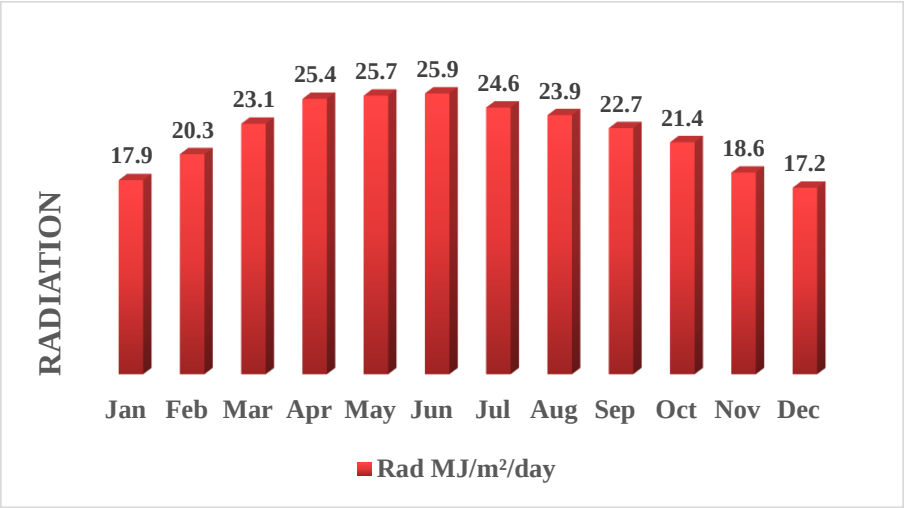

Figure 4.3: Solar Radiation in Abu Hamed

Rainfall in Abu Hamed is scarce (illustrated in Figure 4.4), totaling **12.6 mm** annually, with most precipitation occurring between July and September. The peak rainfall is observed in August, reaching **6.0 mm**, while the majority of the year remains dry. However, effective rainfall ($Reff$), which represents the portion of rainfall available for crop use, is only marginally lower than total rainfall due to minimal losses from runoff or deep percolation. Using the USDA Soil Conservation Service formula, the total annual effective rainfall in Abu Hamed is estimated at 12.5 mm, with the highest effective rainfall occurring in August (**5.9 mm**) and July (**5.1 mm**). Given this low effective rainfall, agricultural production in Abu Hamed remains highly dependent on irrigation, reinforcing the need for efficient and sustainable water management strategies.(S. Ibrahim et al., 2022)

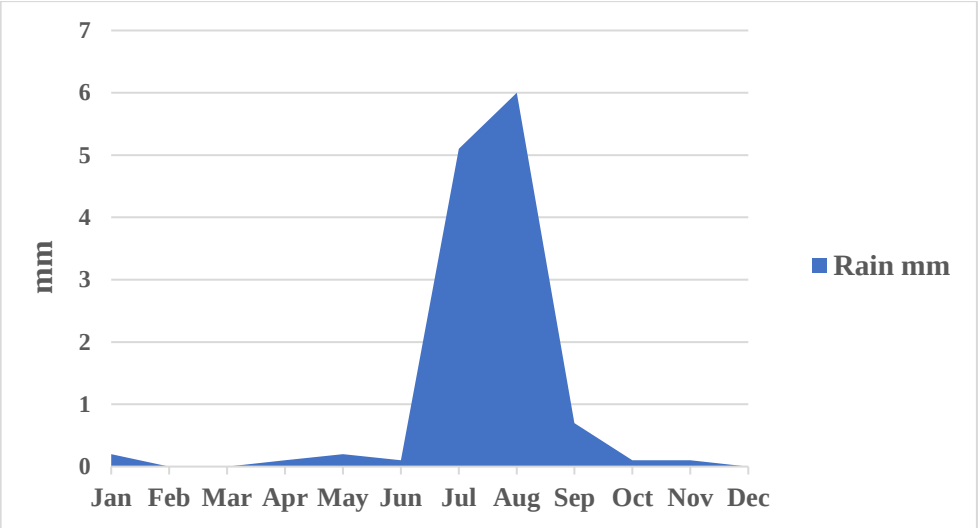

Figure 4.4: Monthly Rainfall in Abu Hamed

4.1.3 Soil Fertility and Irrigation Feasibility

Abu Hamed's agricultural potential is strongly supported by its soil characteristics and access to reliable water sources. The predominant soil type in the region, Red Sandy Loam, demonstrates properties that are conducive to irrigation-based farming. According to data collected from CLIMWAT, this soil type offers the following attributes:

- Total Available Soil Moisture (Field Capacity - Wilting Point): **140.0 mm/meter**, indicating its ability to retain sufficient water to sustain crops during dry periods.
- Maximum Rain Infiltration Rate: **30 mm/day**, which ensures efficient absorption of irrigation water and minimizes surface runoff.
- Maximum Rooting Depth: **900 cm**, allowing crops with deep root systems to access water and nutrients from deeper soil layers.
- Initial Soil Moisture Depletion: **0%**, confirming that the soil is at full capacity at the beginning of the growing season, optimizing conditions for planting.

These properties highlight the soil's suitability for efficient irrigation, particularly in a region with limited rainfall and high evapotranspiration rates.

4.2 Agricultural Practices and Water Requirements

Agriculture in Abu Hamed, located in the River Nile State of Sudan, is a cornerstone of the local economy and a vital source of livelihood for its inhabitants. The region's arid climate, characterized by minimal annual rainfall averaging less than 14 millimeters, necessitates a heavy reliance on irrigation to sustain crop production.

4.2.1 Irrigation Practices

In Abu Hamed, the agricultural sector heavily depends on diesel-powered pumps for irrigation due to the lack of reliable electricity infrastructure. These pumps extract water from the Nile River to irrigate fields, enabling crop cultivation in the region's arid climate.(B. Ali, 2018)

Economic Implications:

The operation of diesel pumps imposes significant financial burdens on farmers. Fluctuating diesel prices, coupled with maintenance costs, can consume a substantial portion of farming expenses. For instance, the cost of diesel in Sudan has been subject to volatility, impacting the predictability of operational expenses for farmers. Additionally, the maintenance of diesel engines requires technical expertise and access to spare parts, which may not be readily available in remote areas, further escalating costs.(Elkadeem et al., 2019)

Environmental Impact:

Diesel-powered pumps contribute to environmental degradation through greenhouse gas emissions and the risk of fuel spills contaminating soil and water sources. Each liter of diesel burned emits approximately 2.68 kilograms of CO_2, exacerbating climate change concerns. Moreover, the noise pollution from diesel engines can disrupt local wildlife and communities.(El-Gafy & El-Bably, 2016)

Operational Challenges:

The reliance on diesel fuel subjects' irrigation practices to the uncertainties of fuel availability and supply chain disruptions. During periods of fuel scarcity, farmers may experience delays in irrigation, adversely affecting crop yields. Furthermore, the manual operation and frequent maintenance of diesel pumps demand considerable time and labor, diverting resources from other essential farming activities.(Elbashir et al., 2024)

4.2.2 Water Requirements

The water requirements for crops in Abu Hamed are substantial, driven by high evapotranspiration rates and prolonged dry seasons. For instance, wheat, a commonly cultivated crop, requires approximately 600 mm per growing season under the region's climatic conditions. Similarly, sorghum requires about 500 mm, while vegetables like tomatoes may demand up to 800 mm.(Khan, Imran, Umer, et al., 2021)

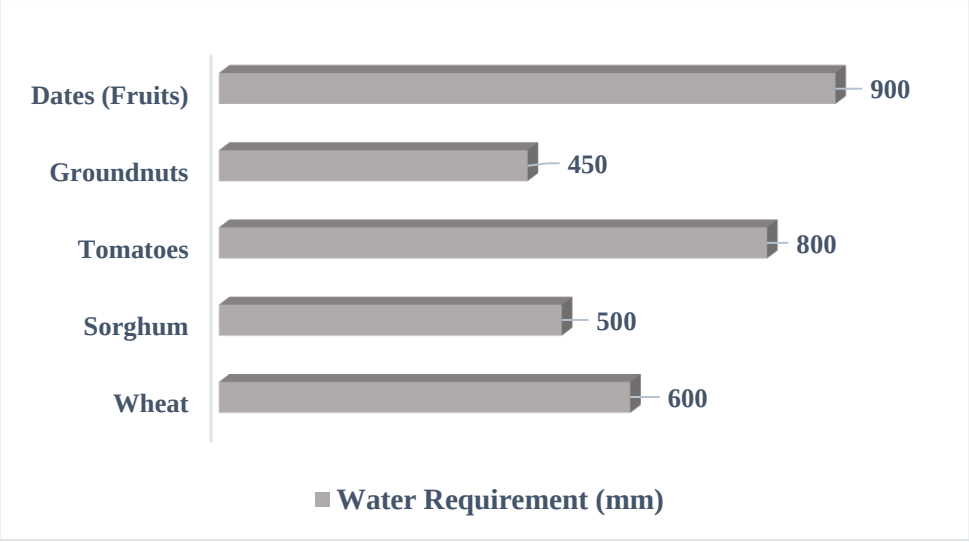

Figure 4.5: Crop Water Requirement in Abu Hamed

These high demands underscore the need for efficient irrigation systems, but the prevalent reliance on surface irrigation methods, which lose up to 50% of water to evaporation and seepage, exacerbates water scarcity. To address these inefficiencies, tailored irrigation scheduling and the adoption of modern techniques such as drip irrigation could optimize water use.(E. Ahmed, 2015)

4.3 Case Study Crop

Wheat has been chosen as the case study crop for this thesis due to its strategic importance to Sudan's economy, its significant irrigation needs, and its adaptability to the agricultural and climatic conditions of Abu Hamed in the River Nile State. As one of Sudan's staple crops, wheat plays a vital role in the country's food security and rural development. This section explores wheat's economic relevance, its high water demands, its suitability for cultivation in Abu Hamed, and the impact of increased production on national food security. Additionally, government initiatives to support wheat cultivation in the River Nile State are examined.

4.3.1 Economic Importance of Wheat

Wheat is a fundamental part of the Sudanese diet, forming the basis of staple foods such as bread and porridge. As a result, it is critical for ensuring food security across the nation. However, Sudan currently imports nearly 70% of its wheat requirements, making it heavily reliant on volatile international markets and foreign exchange reserves. This dependency on imports has exposed the country to supply chain disruptions and price fluctuations, particularly during periods of political or economic instability.

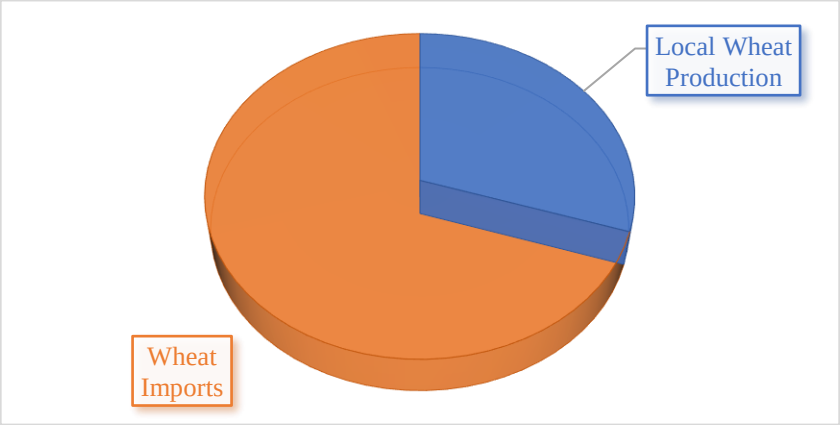

Figure 4.6: Contribution of Wheat to Food Security in Sudan

In Abu Hamed, wheat cultivation provides a crucial source of income for smallholder farmers, who are central to local agricultural production. The crop's economic significance extends beyond farming, creating value along the supply chain, including milling and food production industries. Recognizing its importance, The Sudanese government has actively supported wheat cultivation through initiatives aimed at improving irrigation infrastructure, providing financial incentives, and encouraging the adoption of modern agricultural technologies. In the River Nile State, these efforts have included the

expansion of subsidized access to diesel fuel and irrigation equipment, as well as research programs focused on developing drought-tolerant wheat varieties.(H. A. Ibrahim & Mohamed Baqutayan, 2022; Iizumi et al., 2021)

One noteworthy project involved the rehabilitation of 500 hectares of wheat farmland in Abu Hamed. Supported by international organizations such as the United Nations Development Programme (UNDP), this initiative successfully increased productivity by 25% within three years. These projects highlight the critical role of government and international partnerships in enhancing wheat production and ensuring its sustainability.(H. A. Ibrahim & Mohamed Baqutayan, 2022)

4.3.2 Water Demand

Wheat cultivation in Abu Hamed requires substantial water input due to the region's arid climate and prolonged dry seasons. With annual rainfall averaging less than 14 mm, irrigation is essential to ensure adequate crop growth and yield stability. Studies indicate that wheat typically requires between 450 mm and 650 mm of water per growing season in semi-arid regions, with variations depending on soil conditions, temperature fluctuations, and evapotranspiration rates. In Abu Hamed, this demand is even higher due to the combination of extreme heat, wind intensity, and high evaporation losses from open-field irrigation systems.(H. Ahmed & Mustafa, 2016)

The reliance on diesel-powered pumps for wheat irrigation has led to significant financial and operational constraints for farmers. On average, a hectare of wheat in Abu Hamed requires approximately 5,500–6,500 cubic meters of water per season, necessitating continuous pumping from the Nile River. The energy-intensive nature of diesel pumps results in high fuel consumption, with an estimated 150–180 liters of diesel needed per hectare per irrigation cycle. Rising fuel prices in Sudan have further increased costs, making irrigation expenses unpredictable for wheat farmers.(B. Ali, 2018)

In contrast, transitioning to solar-powered irrigation systems presents a more sustainable and cost-effective solution. Studies conducted in the River Nile State have demonstrated that solar-powered pumps can reduce irrigation costs by up to 60% while ensuring stable water supply, particularly during peak growth periods. Additionally, solar-powered systems enable farmers to optimize irrigation scheduling, reducing water wastage and mitigating the effects of climate variability.(Khan, Imran, Umer, et al., 2021)

4.3.3 Suitability to Abu Hamed

Abu Hamed's agricultural conditions make it a favorable location for wheat cultivation. The region benefits from fertile red sandy loam soils, which provide an ideal medium for wheat root development and nutrient uptake. The proximity of farmland to the Nile River ensures reliable access to irrigation, a critical factor given the region's negligible annual rainfall. Additionally, wheat exhibits a remarkable ability to adapt to Abu Hamed's semi-arid climate, tolerating high temperatures of up to 40°C during its early growth stages.

The region's long growing season, extending from December to August, further supports wheat production by allowing sufficient time for the crop's development and maturation. Combined with the availability of irrigation and the crop's resilience to climate variability, these factors establish Abu Hamed as a strategic area for expanding wheat cultivation.(Iizumi et al., 2021)

4.3.4 Impact on Food Security

Boosting wheat production in Abu Hamed has direct implications for Sudan's food security. Increasing local output reduces reliance on imports, which currently cost the government an estimated $300 million annually in foreign exchange. By stabilizing domestic wheat supplies, the country can mitigate the risks of global market fluctuations, ensuring that affordable wheat-based products remain accessible to its population.(Iizumi et al., 2021)

Enhanced wheat production also contributes to poverty alleviation by creating employment opportunities across the agricultural value chain. For instance, pilot projects introducing modern irrigation technologies have demonstrated a 20% increase in yields, enabling farmers to expand their operations and improve their incomes. These gains underscore the potential for wheat cultivation in Abu Hamed to address both national food security and local economic development goals.(E. Ahmed, 2020)

4.4 Case Study System Modeling

The feasibility of utilizing solar photovoltaic (PV) energy for irrigation was assessed using HOMER Pro, the simulation aimed to compare the feasibility of a solar-powered irrigation system against a diesel-powered alternative, analyzing their respective economic viability, energy sufficiency, and long-term operational sustainability.

System Configuration

Two separate energy systems were modeled:

1. **Solar PV System**
    - **Primary energy source:** Solar PV array
    - **Components:** PV modules, inverter, and AC pump load
    - **Energy supply:** Fully dependent on solar power generation
    - **Economic evaluation:** Based on capital cost, maintenance cost, and levelized cost of energy (LCOE)
    - **Battery storage:** Omitted to simplify system design and reduce capital and maintenance costs. Consequently, the solar system was sized with a larger pump capacity, ensuring the entire irrigation requirement is fulfilled within the available 7 peak sunlight hours each day. This decision guarantees sufficient water pumping capability during daylight hours without energy storage.
2. **Diesel Generator System**
    - **Primary energy source:** Diesel generator
    - **Components:** Diesel engine, fuel supply, and AC pump load
    - **Energy supply:** Fully dependent on fuel consumption
    - **Economic evaluation:** Based on fuel cost, maintenance, and operational expenses over time

The HOMER schematic of the modeled system is presented in **Figure 4.7**, illustrating the interaction between different components.

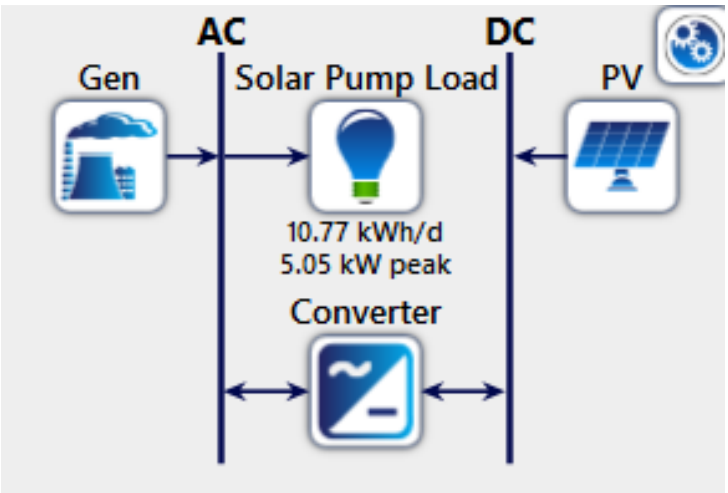

Figure 4.7: Modeled System Schematic

Chapter Five

Results and Discussion

5 Results and Discussion

5.1 Crop Water Requirements

Water availability is a critical factor in determining agricultural productivity, particularly in arid and semi-arid regions such as the River Nile State, Sudan.

based on CROPWAT simulations using local climatic data from Abu-Hamed meteorological station. The analysis includes estimates of evapotranspiration (ETc), net irrigation requirements $NIWR$, and gross irrigation requirements $GIWR$, along with the irrigation scheduling strategy adopted for optimal water use.

5.1.1 Crop Water Demand Analysis

The total water requirement of winter wheat varies throughout the crop growth cycle, with higher water demand occurring during the development and mid-season stages, corresponding to the crop's peak evapotranspiration rates. **Table 5.1** presents a sample of the daily crop water requirements ETc and the effective monthly rainfall.

Month	Growth Stage	Kc	ETc (mm/day)	Effective Rainfall (mm/month)
Dec	Initial	0.7	5.15	0
Jan	Initial	0.72	5.24	0.1
Feb	Development	0.84	7.31	0
Mar	Development	0.94	9.65	0
Apr	Mid-Season	1.09	12.38	0
May	Mid-Season	1.16	13.01	0.35
Jun	Mid-Season	1.22	14.38	0.25
Jul	Late	1.22	13.63	4.95
Aug	Late	0.36	3.78	1.65

Table 5.1: Monthly Crop Water Requirement for Winter Wheat, [CROPWAT Simulation]

From **Table 5.1**, it is evident that water demand peaks between April and June, coinciding with the mid-season growth phase, when evapotranspiration reaches a maximum of **14.38 mm/day**. The total water requirement for winter wheat over the entire growing season is estimated at **2311.5 mm**, with only **7.3 mm** contributed by effective rainfall, reinforcing the necessity for efficient irrigation management.

The detailed crop water needs are shown in **Appendix A**.

5.1.2 Seasonal Distribution of Irrigation Demand

The seasonal water demand profile further illustrates the monthly variations in irrigation needs, as summarized in **Table 5.2.**

Month	Precipitation Deficit (mm)	Net Irrigation Requirement (mm/day)	Net Irrigation Requirement (mm/month)
Jan	168.4	5.4	168.4
Feb	203.6	7.3	203.6
Mar	298.8	9.6	298.8
Apr	356.5	11.9	356.5
May	406.7	13.1	406.7
Jun	425.3	14.2	425.3
Jul	321.2	10.4	321.2
Aug	28.2	0.9	28.2

Table 5.2: Monthly Net Irrigation Requirements for Winter Wheat [CROPWAT SIMULATION]

Water demand is highest from April to June, requiring **over 400 mm** of irrigation water per month.

- July shows a decline in water demand, indicating the crop is entering the late-season phase.
- Effective rainfall contributions are negligible, necessitating full reliance on irrigation for optimal crop growth.

The detailed seasonal distribution is shown in **Appendix B**

5.1.3 Seasonal Irrigation Scheduling & Load Profile

Efficient irrigation scheduling is essential for optimizing water use efficiency and ensuring a continuous energy supply for pumping operations. The irrigation schedule was developed to match the crop water requirement (CWR) while minimizing water losses and excessive pumping demand. **Table 5.3** provides an insight of the irrigation scheduling strategy, then we analyze the seasonal load profile, ensuring alignment between energy demand and water availability.

Irrigation Scheduling Strategy

The irrigation schedule was designed considering:

- Refilling soil moisture to 100% of field capacity at each irrigation event.
- Operating at full pump capacity during peak irrigation demand periods.
- Applying irrigation at predefined intervals, accounting for crop growth stages and soil moisture depletion.

Date	Growth Stage	Net Irrigation (mm)	Gross Irrigation (mm)	Flow Rate (l/s/ha)
17-Dec	Initial	30.9	44.1	0.85
30-Dec	Initial	36	51.5	0.85
7-Jan	Initial	41	58.6	0.85
12-Feb	Development	60.9	87.1	1.12
21-Feb	Development	66.5	95	1.22
2-Mar	Development	73.9	105.6	1.36
28-Apr	Development	99	141.4	2.05
16-May	Development	116	165.7	2.13
3-Jun	Mid-Season	124.3	177.5	2.28
21-Jun	Mid-Season	129.1	184.4	2.37
9-Jul	Mid-Season	121.4	173.4	2.23
8-Aug	End-Season	0	0	0

Table 5.3: Irrigation Schedule for Winter Wheat [CROPWAT Simulation]

The results confirm that irrigation events are more frequent during the mid-season growth phase, aligning with the highest crop water demand recorded in April, May, and June. During these months, the pumping system must sustain continuous operation, ensuring that irrigation requirements are met without compromising soil moisture levels or crop growth.

The detailed irrigation schedule is shown in **Appendix C**

Load Profile Analysis

To evaluate the seasonal energy demand for irrigation, the pump's peak load and daily energy consumption were used to generate a monthly load profile. Given the absence of precise hourly irrigation scheduling data, the pump was modeled as a constant daily load, meaning that:

- HOMER assumes a uniform energy consumption throughout the day, simplifying system modeling.
- Peak demand periods within the day were not explicitly simulated, as the load was averaged over the operating hours.

While these simplifications do not capture variations in daily pumping schedules, it provides a simplified yet conservative estimate of the system’s energy demand, ensuring that the irrigation system remains functional throughout the day without interruptions.

The estimated AC primary load derived from HOMER is presented in **Figure 5.1**, illustrating the monthly average energy demand.

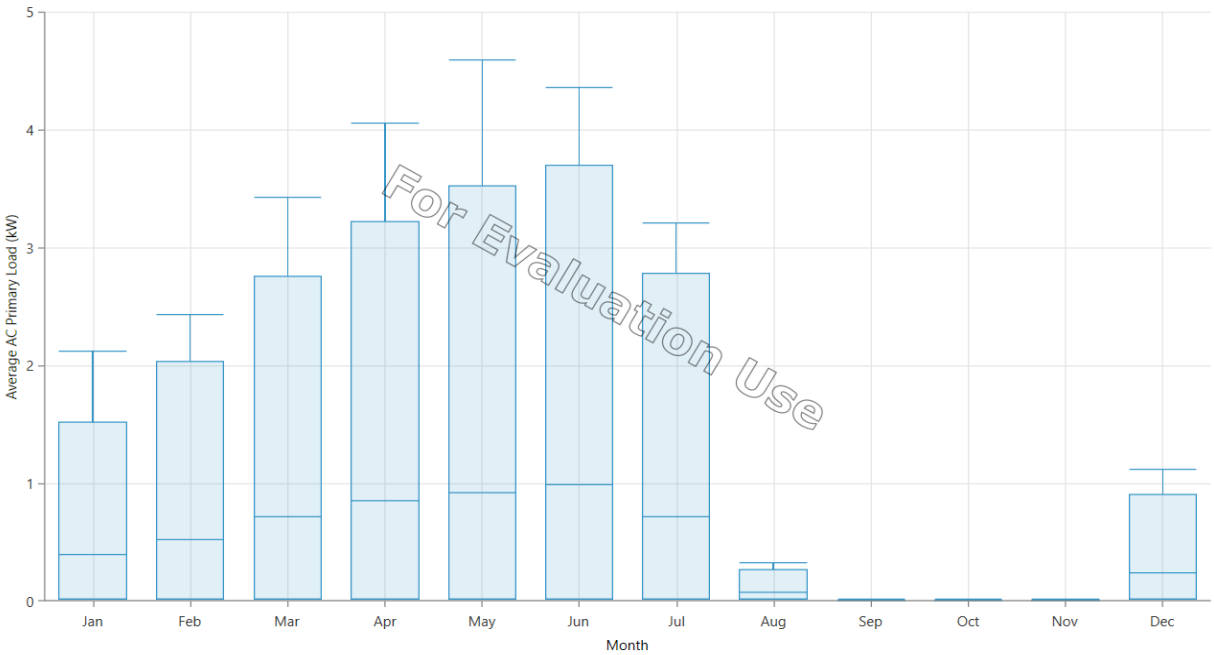

Figure 5.1: AC primary load monthly averages [Homer Pro Simulation]

The load profile closely follows the seasonal water demand pattern, confirming that:

- Peak energy consumption occurs during high irrigation months (April–June).
- Energy demand gradually declines in the late season, reflecting the reduction in irrigation frequency.

In addition to the monthly energy consumption trends, the daily load profile of the irrigation system provides further insight into the operational pattern of the pump.

Figure 5.2 presents the AC primary load daily profile, illustrating the typical power demand of the system across different months.

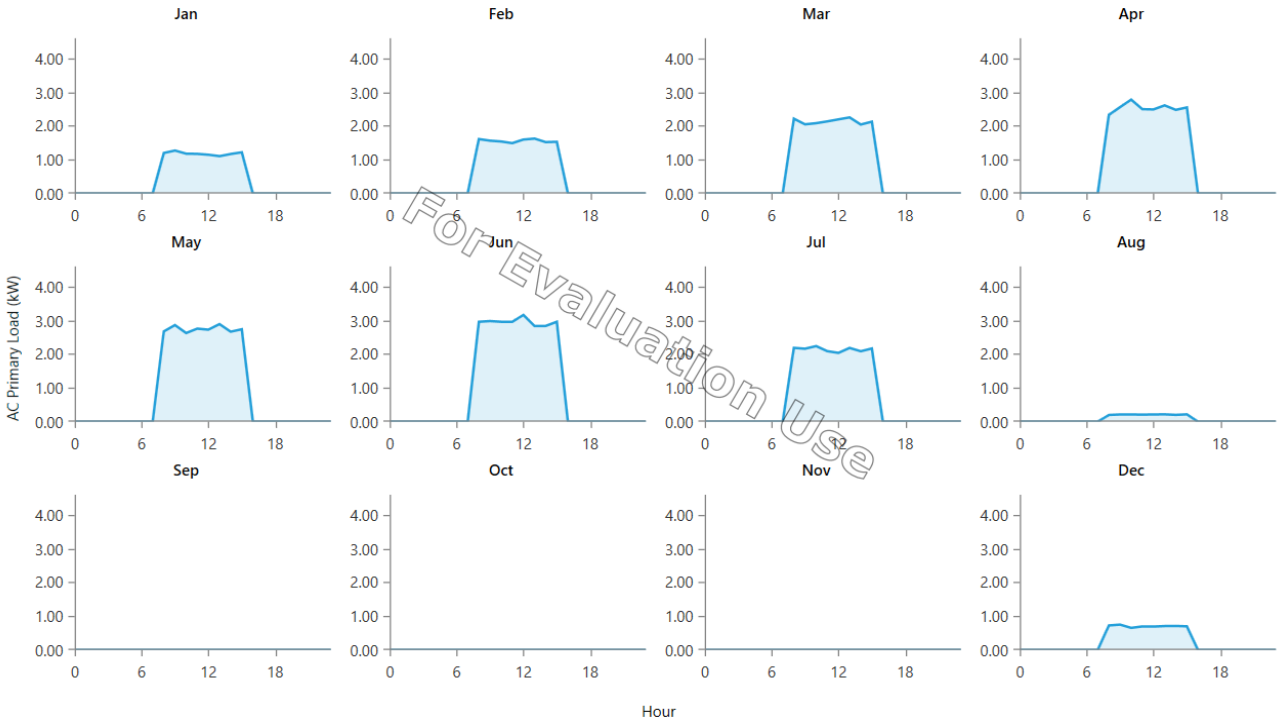

Figure 5.2:AC primary load daily profile [Homer Pro Simulation]

The daily load profile reveals that the pumping system consistently operates between 8:00 AM and 3:00 PM, with no energy consumption outside these hours. This behavior is observed across the months with irrigation demand, particularly between December to August, while in the remaining months, the load is absent, reflecting the seasonal variation in irrigation needs.

It is evident that the peak power demand varies slightly between months, with higher loads observed in April through June, coinciding with the months of higher irrigation water requirements. Additionally, the shape of the curves suggests that the pump operates at nearly full capacity during the entire active period, ensuring that water is delivered efficiently without interruptions.

This operational pattern aligns with the solar-based system design, which was deliberately configured to function without battery storage. Since solar energy generation follows a similar daily cycle—rising in the morning, peaking around midday, and declining in the afternoon—the system ensures that irrigation pumping takes place only when solar power is available. By restricting pump operation to daylight hours, the need for costly battery storage is eliminated, simplifying the system while maintaining reliability, while this configuration is simpler it can reduce the performance and crop resilience of the system

because it does not account for weather variability and other natural disruptors that can be mitigated by battery usage.

5.2 Solar PV System Performance and Capacity

The solar photovoltaic (PV) system was optimized in HOMER Pro to supply the irrigation energy demand efficiently while maintaining a fully renewable energy supply. The simulation determined the required PV system capacity, inverter size, and total energy production, ensuring reliable operation without the need for additional energy storage.

Installed PV Capacity and Energy Output

The HOMER simulation identified an optimal PV system capacity of **30.59 kW**, generating an estimated **57.89 MWh** per year to meet the irrigation load. The system operates with a renewable energy fraction of 100%, ensuring that all energy required for water pumping is supplied directly from solar power without reliance on fossil fuels.

The monthly electricity production is illustrated in **Figure 5.3**, demonstrating a relatively stable power generation pattern throughout the year. The system produces between **3.8 MWh** and **5.2 MWh** per month, with peak production occurring in March to October, reaching approximately **5.2 MWh**. The lowest generation is observed in December, where PV output decreases to around **3.8 MWh** due to reduced solar radiation.

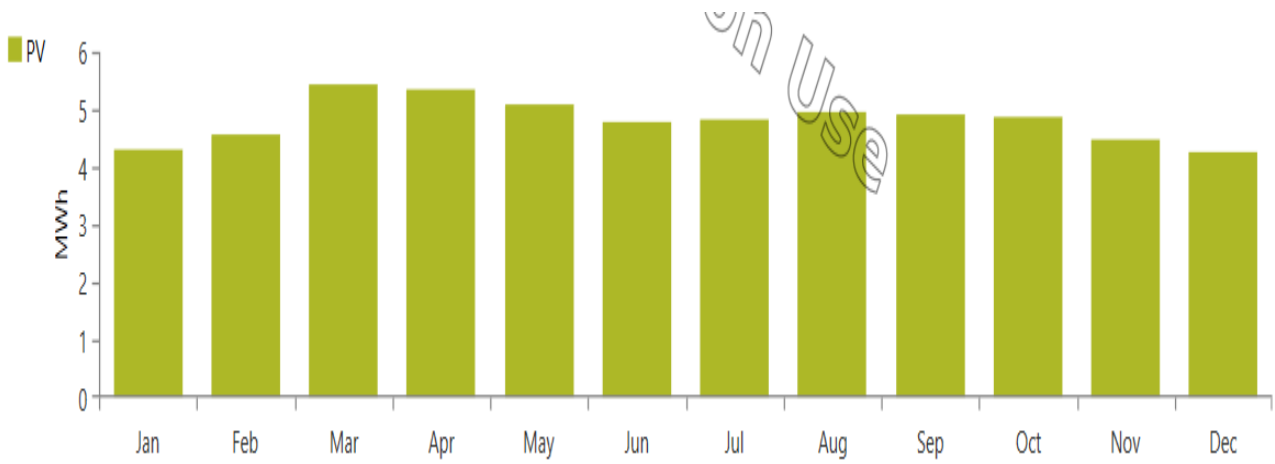

Figure 5.3: Monthly Electric production [Homer Pro Simulation]

Despite these seasonal variations, solar PV consistently generates sufficient energy to power the irrigation system, with no significant shortfalls requiring backup energy sources. The monthly average PV power output, shown in **Figure 5.4**, further illustrates the solar energy availability throughout different months.

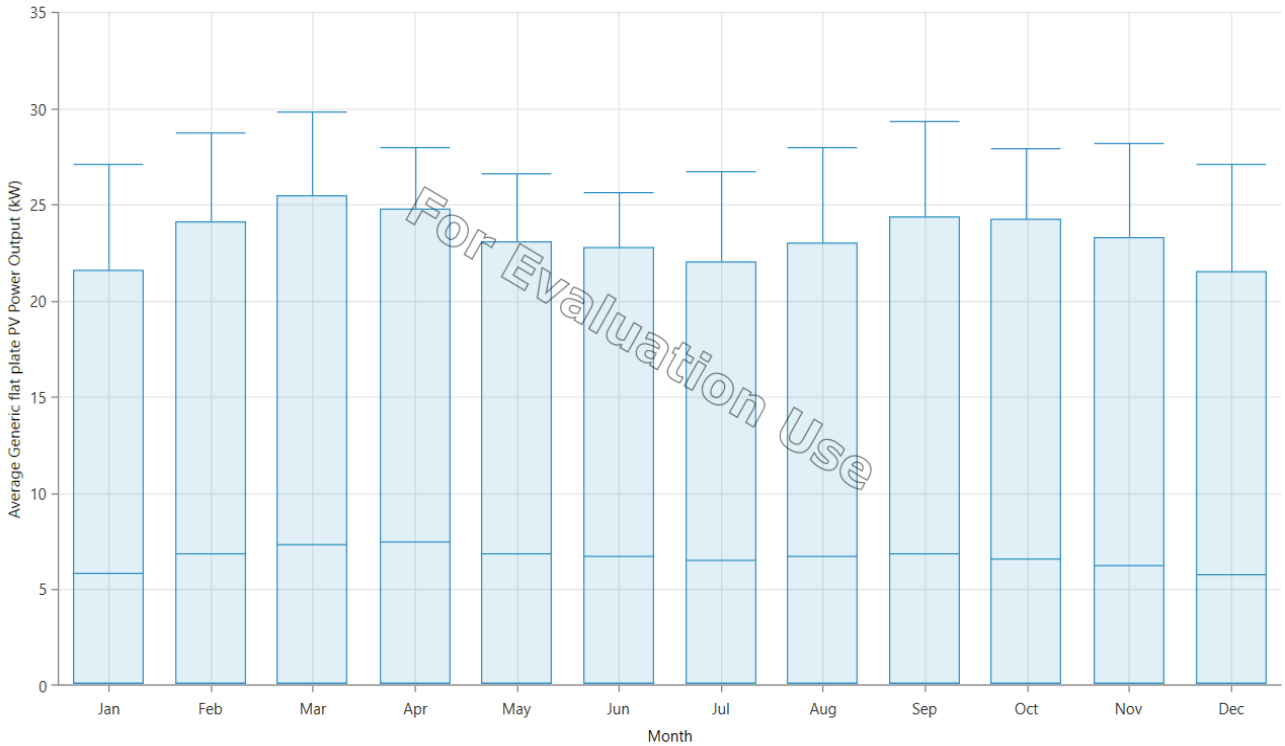

Figure 5.4: Generic flat plate PV Power Output Monthly Averages [Homer Pro Simulation]

The system's average monthly power output ranges between 20 kW and 30 kW, peaking in months of high solar irradiance. The error bars indicate fluctuations in daily power production, likely influenced by cloud cover and seasonal solar variations. However, the consistency in PV generation confirms the feasibility of the system to operate without requiring battery storage, as irrigation pumping is scheduled to match solar availability.

System Components' Specifications

The solar-powered irrigation system consists of the following key components:

- **PV Array: 30.59 kW**, serving as the primary power source.
- **Inverter: 5.5 kW**, converting DC power from PV panels into AC power for the pump.

Operating the pump exclusively during daylight hours eliminates the need for battery storage, reducing capital and operational costs while maximizing direct solar energy utilization. This design ensures a cost-effective and sustainable irrigation solution by directly linking energy production to water pumping requirements.

5.3 Economic Comparison

The financial feasibility of the solar-powered irrigation system was evaluated against a diesel-powered alternative, considering the total lifetime cost, cost per unit of electricity, and operational expenses. The comparison accounts for Net Present Cost NPC, Levelized Cost of Energy $LCOE$, initial capital investment, operating and maintenance (O&M) costs, and long-term financial sustainability.

The Net Present Cost NPC, which is the present value of all the costs of installing and operating the Component over the project lifetime, minus the present value of all the revenues that it earns over the project lifetime, highlights a clear cost disparity between the two systems. As seen in **Table 5.4**, the NPC for the solar system is **$12,890**, significantly lower than the diesel system's **$45,177**. The difference is mainly due to fuel and maintenance costs associated with diesel generators, which accumulate significantly over the system's lifespan. While the initial capital cost for solar PV is higher **($11,791)** than diesel **($2,800)**, the low ongoing expenses ensure a much lower total cost over the 25-year analysis period.

Cost Summary

	Base Case	Lowest Cost System
NPC	$45,177	$12,890
Initial Capital	$2,800	$11,791
O&M	$3,278/yr	$84.98/yr
LCOE	$0.889/kWh	$0.254/kWh

Table 5.4: Economic Comparison [Homer Pro Simulation]

The Levelized Cost of Energy $LCOE$, which is a measure of the average net present cost of electricity generation over its lifetime, measures the cost per kWh of electricity generated, further illustrates the economic advantage of solar energy.

As shown in **Table 5.4**, the solar system achieves an $LCOE$ of **$0.254/kWh**, significantly lower than the diesel system's **$0.889/kWh**. This confirms that, despite its higher upfront investment, solar energy provides a more cost-efficient power source in the long run, as it does not require continuous fuel expenditures.

The operating and maintenance (O&M) costs further emphasize this cost advantage. The solar system incurs minimal annual expenses, with O&M costs estimated at just **$84.98** per year, primarily for cleaning and minor component inspections. In contrast, the diesel system requires **$3,278** per year, driven by frequent maintenance, engine servicing, and fuel replenishment. These ongoing costs significantly contribute to the high NPC of diesel-based irrigation.

The cumulative cash flow analysis, illustrated in **Figure 5.5**, provides insight into how costs accumulate over time. The diesel system follows a steep downward trajectory, reflecting the continuous spending on fuel and maintenance, which steadily increases its financial burden. By the 10th year, the diesel system has already incurred costs exceeding **$30,000**, and by the end of the **25-year period**, total expenditures approach **$90,000**.

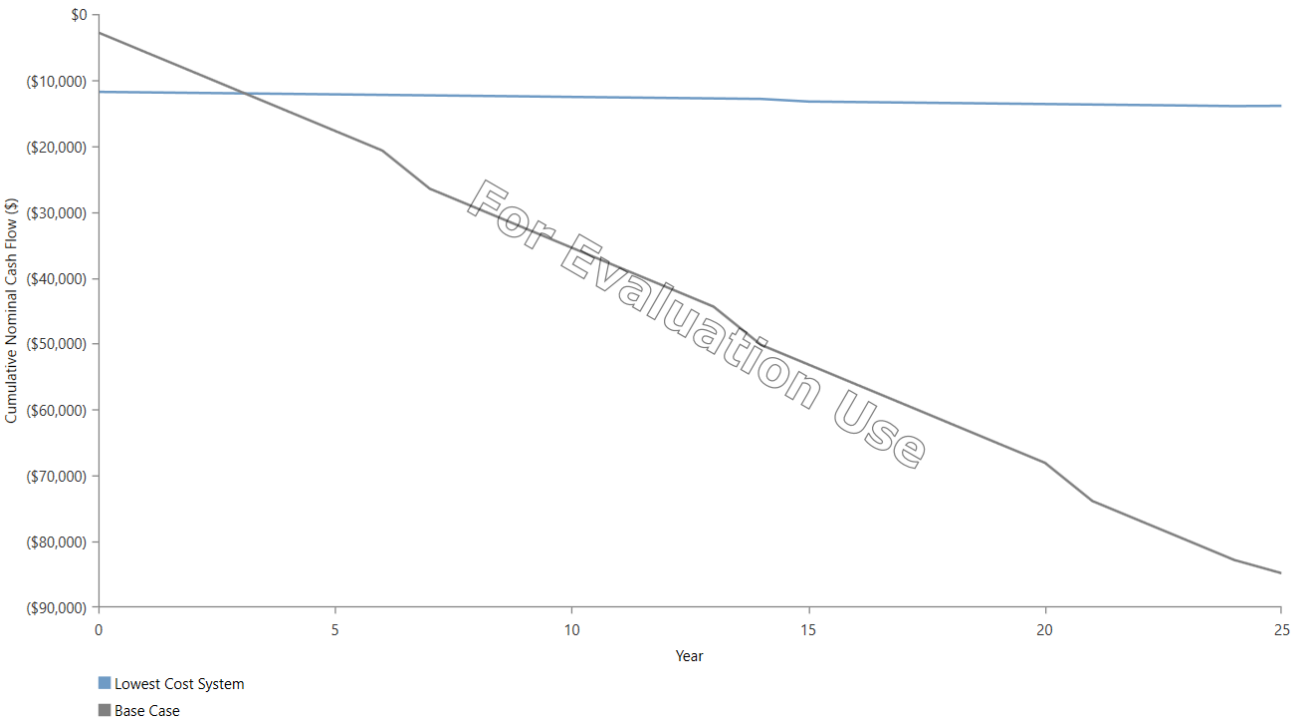

Figure 5.5: Cumulative Nominal Cash Flow [Homer Pro Simulation]

On the other hand, the solar system maintains a nearly flat cost trajectory, as it requires only an initial investment and negligible annual expenses. This cost stability ensures that, after the payback period of **3.09 years**, all subsequent energy generation is effectively free, leading to substantial long-term savings.

Further supporting this investment, the Internal Rate of Return (IRR) for the solar system is **33.86%**, indicating a highly attractive return on investment. This is particularly relevant in Abu Hamed where the fuel price fluctuations and supply chain issues make diesel systems even more costly and unreliable.

The cash flow breakdown in **Figure 5.6** further highlights the stark contrast between the two systems. The solar system incurs a single large expense at the beginning, followed by nearly zero recurring costs, while the diesel system experiences frequent, repeated expenses every year, making it financially unsustainable in the long term.

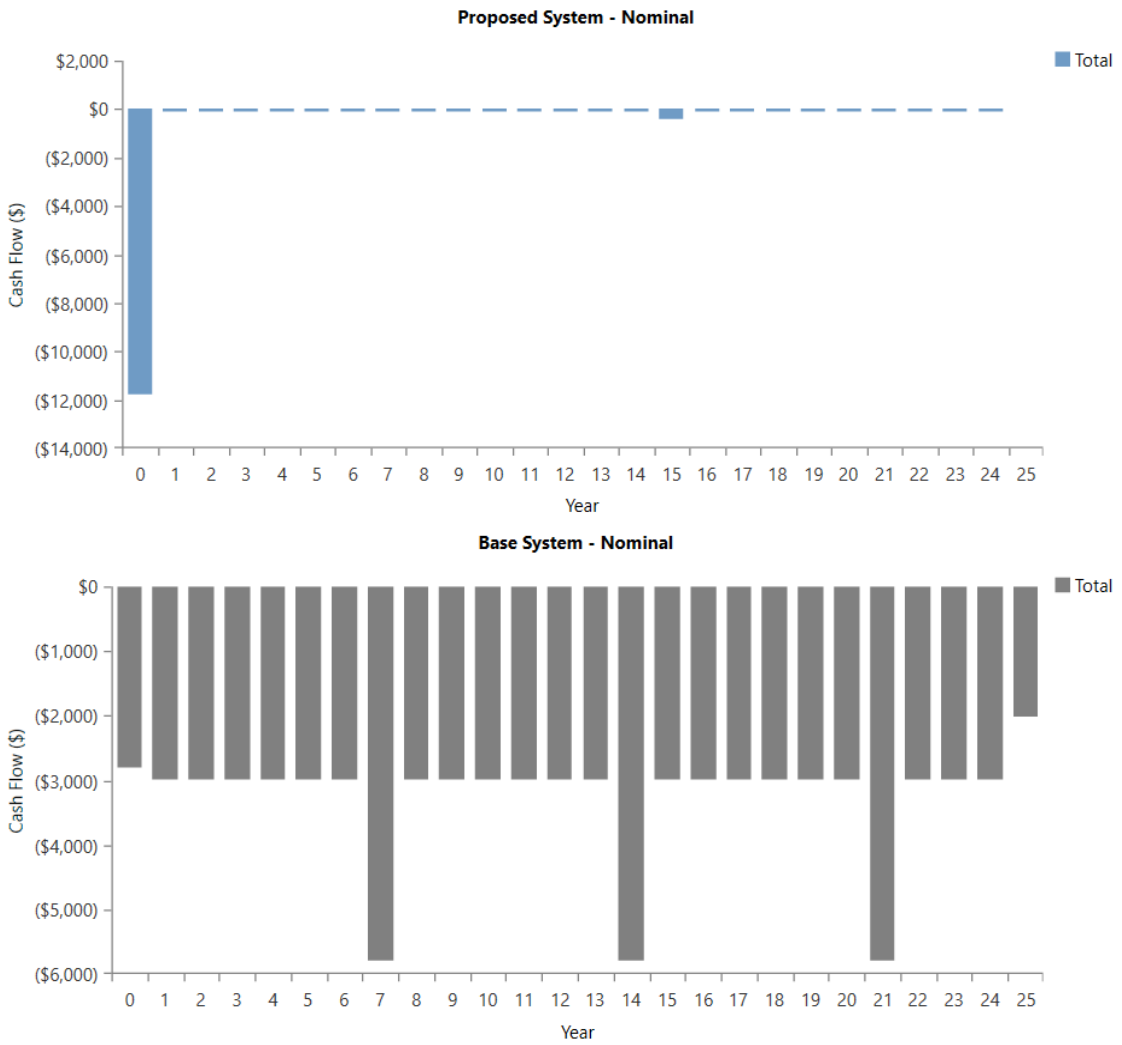

Figure 5.6: Cash Flow Breakdown [Homer Pro Simulation]

5.4 Operational and Practical Considerations

While the economic analysis demonstrates the financial advantages of solar-powered irrigation over diesel systems, the implementation of solar-powered irrigation, despite its economic advantages, requires consideration of several practical and operational factors. These include system reliability, maintenance requirements, external risks, and the potential for future expansion.

System Reliability and Performance

Ensuring a stable and sufficient energy supply is a key factor in the feasibility of solar irrigation. As demonstrated in **Figure 5.3**, solar power generation remains relatively stable throughout the year, with monthly production varying between 3.8 MWh and 5.2 MWh. Although minor seasonal fluctuations exist, they align well with irrigation demand, confirming that the system can provide adequate power without major interruptions.

Unlike diesel generators, which provide power on demand, solar-powered irrigation depends on daylight availability. However, since the irrigation schedule is designed to operate within solar production hours, this dependency does not present a significant operational challenge. The decision to eliminate battery storage simplifies the system while reducing costs, as all generated energy is used directly to meet pumping requirements.

Maintenance and Technical Requirements

Solar PV systems require minimal maintenance compared to diesel generators. The main maintenance activities include regular cleaning of the PV panels to prevent dust accumulation, periodic inspections of electrical connections, and monitoring of inverter performance. These tasks ensure the system continues to operate at optimal efficiency with minimal downtime.

Diesel-powered systems, in contrast, demand frequent maintenance, including oil changes, filter replacements, and mechanical servicing, all of which contribute to their high annual operating costs. Additionally, diesel engines are more prone to wear and tear due to their moving mechanical components, whereas solar PV systems operate without significant degradation over time.

Despite the advantages of low-maintenance solar systems, one potential challenge is the availability of skilled technicians for installation and servicing. In remote agricultural regions like Abu Hamed, limited technical expertise can pose a barrier to long-term system

sustainability. Addressing this challenge through training programs and local capacity-building efforts would help ensure the continued operation of solar irrigation systems.

External Risks and Long-Term Viability

Several external factors could affect the long-term feasibility of both solar and diesel systems. The price volatility of diesel fuel is a major financial risk for diesel-powered irrigation, as fuel costs are subject to market fluctuations, geopolitical influences, and supply chain disruptions. In contrast, solar-powered irrigation eliminates fuel dependency, providing a stable and predictable cost structure.

Climate conditions also play a role in system performance. While solar radiation levels are generally consistent in Abu Hamed, extreme weather events or prolonged cloudy periods could temporarily reduce energy generation. However, given the alignment between peak irrigation demand and high solar availability, this risk remains minimal.

Component longevity is another factor to consider. While solar panels typically last over 25 years, inverters and electrical components may require replacement within 10 to 15 years. Proper planning for component replacement costs is necessary to ensure the long-term sustainability of the system.

Scalability and Future Expansion

One of the key advantages of solar irrigation is its scalability. As irrigation demand increases due to expanding cultivated areas or changing climate conditions, additional PV panels can be integrated to increase system capacity. Unlike diesel systems, where expansion would require larger fuel storage and increased operational costs, scaling up a solar irrigation system can be done incrementally without significant long-term financial burdens.

Additionally, integrating the solar system with water storage solutions can further enhance its efficiency. By using excess solar energy to pump and store water for later use, farmers can ensure continuous irrigation even during periods of reduced sunlight, improving the system’s resilience.

By addressing these operational and practical considerations, solar-powered irrigation can be implemented as a sustainable and cost-effective alternative to diesel-based systems.

5.5 Environmental Impact of Solar and Diesel Irrigation

The transition from diesel-powered to solar-powered irrigation provides substantial environmental benefits as well. Diesel generators contribute to carbon dioxide (CO_2) emissions, air pollution, and fuel-related environmental hazards, whereas solar-powered systems operate with zero direct emissions. This section evaluates the environmental impact of both systems, highlighting the potential reductions in greenhouse gas (GHG) emissions achieved through solar irrigation.

Carbon Emissions from Diesel-Powered Irrigation

Diesel-based irrigation systems rely on continuous fuel consumption, leading to substantial CO_2 emissions over time. The generator used in this study, with a rated capacity of **5.6 kW**, operates for approximately **2,192 hours** per year to meet irrigation energy demands. Taking an estimated fuel consumption rate of **0.4 liters per kWh**, this results in an annual fuel consumption of approximately **1743.5 liters of diesel**.

With each liter of diesel emitting **2.68 kg of CO_2** during combustion, the total annual emissions from the diesel-powered irrigation system amount to approximately **4.67 metric tons of CO_2 per year**. This highlights the significant carbon footprint associated with conventional irrigation methods, particularly in regions like Abu Hamed where diesel remains the primary energy source for agricultural applications.

Environmental Benefits of Solar-Powered Irrigation

In contrast, the solar PV system operates without any fuel consumption, resulting in zero direct emissions throughout its lifetime. By replacing diesel-based irrigation with a 30.59 kW solar system, an estimated 4.67 metric tons of CO_2 emissions are avoided annually. Over a 25-year system lifespan, this translates to a cumulative reduction of over 116 metric tons of CO_2, reinforcing the long-term sustainability benefits of solar energy.

Beyond carbon emissions, diesel-powered systems contribute to air pollution through nitrogen oxides (NO_x) and particulate matter (PM), which degrade air quality and pose health risks to farm workers. Additionally, diesel storage and transportation present environmental hazards, as fuel spills can lead to soil and water contamination. By eliminating fuel dependency, solar irrigation mitigates these risks, providing a cleaner and safer alternative for agricultural operations.

Long-Term Sustainability Considerations

The adoption of solar-powered irrigation supports global efforts to transition away from fossil fuels, aligning with climate policies aimed at reducing agricultural carbon footprints. With an operational lifespan of over 25 years, solar PV systems offset emissions while eliminating the financial and environmental uncertainties associated with fuel-based irrigation. Although PV panel manufacturing involves some embedded emissions, these are significantly lower than the cumulative emissions produced by diesel systems over the same timeframe.

Furthermore, ongoing advancements in solar panel recycling and energy storage continue to enhance the sustainability of solar technology. As renewable energy adoption accelerates, solar irrigation represents a future-proof solution for sustainable water management in agriculture, ensuring both economic viability and environmental responsibility.

Chapter Six

Conclusion

6 Conclusion

This study investigated the viability of solar-powered water pumping systems as a sustainable alternative to diesel-powered irrigation for small-scale farmers in Sudan's River Nile State. The study used CROPWAT for technical analysis and HOMER Pro for economic modelling to evaluate the water requirements for wheat cultivation and compared the operational, environmental, and financial performance of solar and diesel irrigation systems under site-specific conditions.

The findings clearly indicate that solar pumping systems is a technically feasible and economically advantageous solution in the long term, especially in regions with high solar irradiance and unreliable access to diesel. The results show that while solar systems entail a higher initial investment, they offer a **60%** reduction in operating costs and more than **3.5 tons of CO_2** savings per season compared to diesel-powered systems due to the elimination of fuel consumptions and lower maintenance requirements. Moreover, solar pumping systems carry environmental benefits by reducing emissions and local pollution, contributing to Sudan's sustainability goals.

However, the study also highlights critical barriers to widespread adoption. The high upfront cost of solar pumping systems, lack of financing options, limited technical knowledge among farmers, and absence of institutional support remain significant challenges. These findings support the thesis argument that external financial and educational assistance is essential to make solar irrigation accessible to rural farmers. Without such support, the technology's potential may remain unrealized in regions where it is most needed.

By combining agro-climatic and economic analysis within a real-world Sudanese context, this research contributes new empirical data to the growing literature on renewable energy in agriculture. It reinforces the need for integrated planning that connects energy, water, and food systems with development policy. For practitioners and policymakers, the study offers a replicable modeling approach and provides evidence-based insights that can inform financing mechanisms, capacity-building programs, and national renewable energy strategies.

Future research could build on this work by incorporating more diverse crop types, modeling battery or hybrid system performance, and exploring community-based financing models. Longitudinal field studies are also needed to evaluate system performance under actual operational conditions. Ultimately, enabling the transition to solar-powered irrigation in Sudan will require coordinated efforts across technical, financial, and institutional dimensions, but doing so holds great promise for promoting sustainable agriculture and rural development.

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Appendix

Appendix A: Wheat Water Requirements Output (CROPWAT)

This appendix presents the detailed results of the crop water requirement estimation for winter wheat in Abu Hamed, River Nile State, as generated using CROPWAT 8.0.

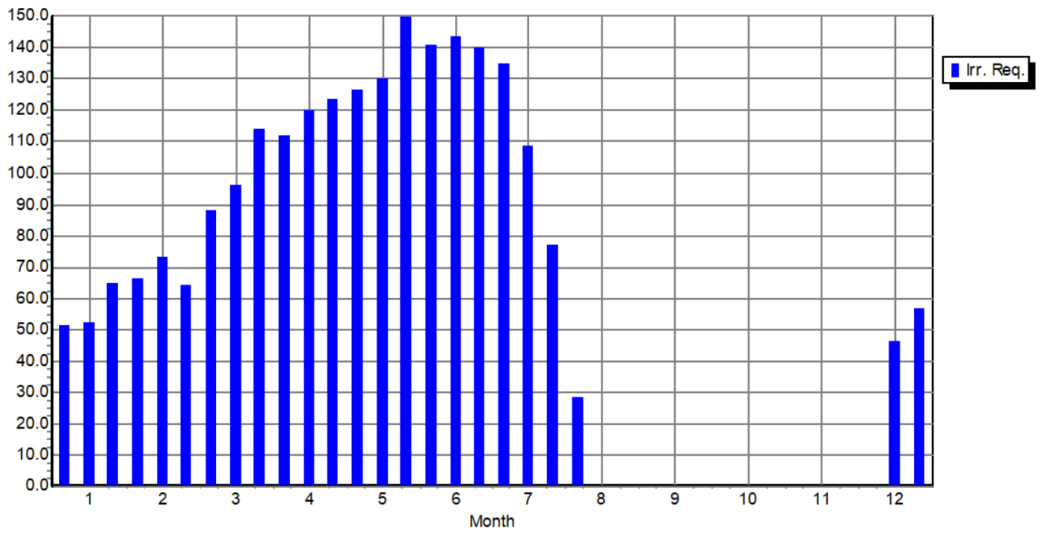
CROP WATER REQUIREMENTS

ETo station: ABU-HAMED
Rain station: ABU-HAMED

Crop: Winter Wheat f.f.
Planting date: 12/12

Month	Decade	Stage	Kc coeff	ETc mm/day	ETc mm/dec	Eff rain mm/dec	Irr. Req. mm/dec
Dec	2	Init	0.70	5.15	46.3	0.0	46.3
Dec	3	Init	0.70	5.15	56.6	0.0	56.6
Jan	1	Init	0.70	5.13	51.3	0.0	51.3
Jan	2	Deve	0.72	5.24	52.4	0.1	52.3
Jan	3	Deve	0.76	5.90	64.8	0.0	64.8
Feb	1	Deve	0.80	6.62	66.2	0.0	66.2
Feb	2	Deve	0.84	7.31	73.1	0.0	73.1
Feb	3	Deve	0.87	8.04	64.3	0.0	64.3
Mar	1	Deve	0.90	8.81	88.1	0.0	88.1
Mar	2	Deve	0.94	9.65	96.5	0.0	96.5
Mar	3	Deve	0.98	10.39	114.3	0.0	114.3
Apr	1	Deve	1.02	11.23	112.3	0.0	112.3
Apr	2	Deve	1.05	12.05	120.5	0.0	120.5
Apr	3	Deve	1.09	12.38	123.8	0.0	123.7
May	1	Deve	1.13	12.66	126.6	0.1	126.5
May	2	Deve	1.16	13.01	130.1	0.1	130.0
May	3	Mid	1.20	13.65	150.2	0.1	150.1
Jun	1	Mid	1.22	14.12	141.2	0.0	141.2
Jun	2	Mid	1.22	14.38	143.8	0.0	143.8
Jun	3	Mid	1.22	14.04	140.4	0.2	140.2
Jul	1	Late	1.22	13.63	136.3	1.2	135.0
Jul	2	Late	1.01	11.06	110.6	1.8	108.8
Jul	3	Late	0.67	7.20	79.3	1.9	77.4
Aug	1	Late	0.36	3.78	30.2	1.6	28.2
					2319.3	7.3	2311.5

Appendix A-1: Crop water requirement table



Appendix A-2: Crop water requirement chart

Appendix B: Scheme Output (CROPWAT)

This appendix summarizes monthly net scheme irrigation requirements and precipitation deficits across the wheat growing season using CROPWAT 8.0. The data illustrates seasonal irrigation demands, average daily and monthly flow requirements, and coverage rates for irrigated areas.

SCHEME SUPPLY

ETo station: ABU-HAMED Cropping pattern: dageegna
Rain station: ABU-HAMED

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Precipitation deficit												
1. Winter Wheat f.f.	168.4	203.6	298.8	356.5	406.7	425.3	321.2	28.2	0.0	0.0	0.0	102.9
Net scheme irr.req.												
in mm/day	5.4	7.3	9.6	11.9	13.1	14.2	10.4	0.9	0.0	0.0	0.0	3.3
in mm/month	168.4	203.6	298.8	356.5	406.7	425.3	321.2	28.2	0.0	0.0	0.0	102.9
in l/s/h	0.63	0.84	1.12	1.38	1.52	1.64	1.20	0.11	0.00	0.00	0.00	0.38
Irrigated area (% of total area)	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	0.0	0.0	0.0	100.0
Irr.req. for actual area (l/s/h)	0.63	0.84	1.12	1.38	1.52	1.64	1.20	0.11	0.00	0.00	0.00	0.38

Appendix B: Scheme table

Appendix C: Irrigation Schedule Output (CROPWAT)

Detailed irrigation schedule using CROPWAT 8.0 is used for simulation, indicating application dates, net and gross irrigation depths, field efficiency, and corresponding flow rates based on 100% soil moisture depletion and full field capacity refill.

CROP IRRIGATION SCHEDULE

ETo station: ABU-HAMED Crop: Winter Wheat f.f. Planting date: 12/12
Rain station: ABU-HAMED Soil: RED SANDY LOAM Harvest date: 08/08

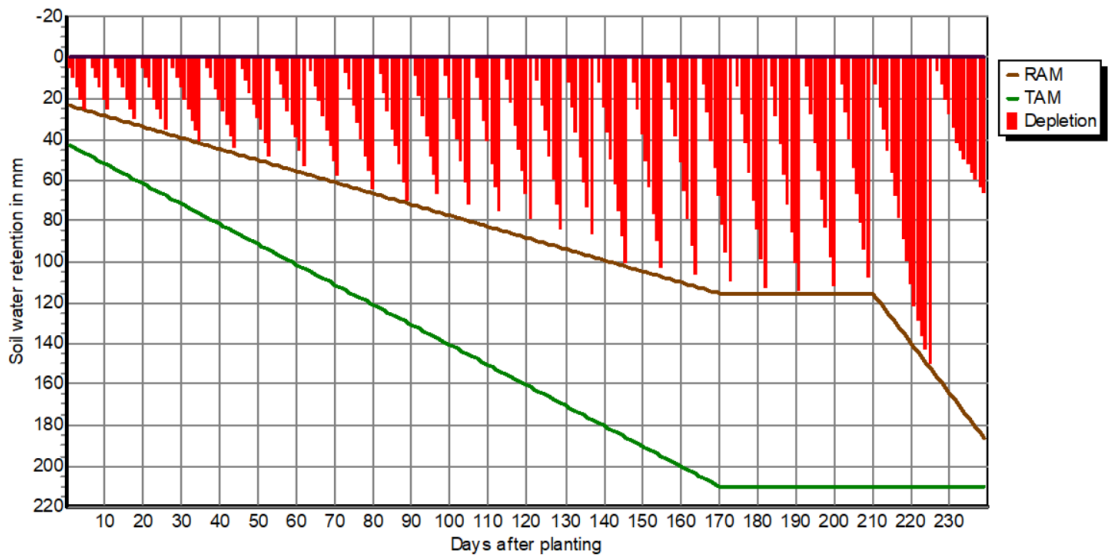
Yield red.: 0.0 %

Crop scheduling options
Timing: Irrigate at 100 % depletion
Application: Refill to 100 % of field capacity
Field eff. 70 %

Table format: Irrigation schedule

Date	Day	Stage	Rain mm	Ks fract.	Eta %	Depl %	Net Irr mm	Deficit mm	Loss mm	Gr. Irr mm	Flow l/s/ha
17 Dec	6	Init	0.0	1.00	100	64	30.9	0.0	0.0	44.1	0.85
23 Dec	12	Init	0.0	1.00	100	57	30.9	0.0	0.0	44.1	0.85
30 Dec	19	Init	0.0	1.00	100	59	36.0	0.0	0.0	51.5	0.85
7 Jan	27	Init	0.0	1.00	100	60	41.0	0.0	0.0	58.6	0.85
16 Jan	36	Dev	0.0	1.00	100	60	46.8	0.0	0.0	66.8	0.86
25 Jan	45	Dev	0.0	1.00	100	58	50.4	0.0	0.0	72.0	0.93
3 Feb	54	Dev	0.0	1.00	100	58	55.2	0.0	0.0	78.9	1.01
12 Feb	63	Dev	0.0	1.00	100	58	60.9	0.0	0.0	87.1	1.12
21 Feb	72	Dev	0.0	1.00	100	59	66.5	0.0	0.0	95.0	1.22
2 Mar	81	Dev	0.0	1.00	100	61	73.9	0.0	0.0	105.6	1.36
11 Mar	90	Dev	0.0	1.00	100	61	80.1	0.0	0.0	114.4	1.47
19 Mar	98	Dev	0.0	1.00	100	56	77.2	0.0	0.0	110.2	1.59
27 Mar	106	Dev	0.0	1.00	100	56	82.4	0.0	0.0	117.7	1.70
4 Apr	114	Dev	0.0	1.00	100	56	86.5	0.0	0.0	123.5	1.79
12 Apr	122	Dev	0.0	1.00	100	56	91.5	0.0	0.0	130.7	1.89
20 Apr	130	Dev	0.0	1.00	100	57	96.4	0.0	0.0	137.7	1.99
28 Apr	138	Dev	0.0	1.00	100	55	99.0	0.0	0.0	141.4	2.05
7 May	147	Dev	0.0	1.00	100	60	113.3	0.0	0.0	161.9	2.08
16 May	156	Dev	0.0	1.00	100	59	116.0	0.0	0.0	165.7	2.13
25 May	165	Dev	0.0	1.00	100	59	120.3	0.0	0.0	171.8	2.21
3 Jun	174	Mid	0.0	1.00	100	59	124.3	0.0	0.0	177.5	2.28
12 Jun	183	Mid	0.0	1.00	100	61	127.6	0.0	0.0	182.3	2.34
21 Jun	192	Mid	0.0	1.00	100	61	129.1	0.0	0.0	184.4	2.37
30 Jun	201	Mid	0.0	1.00	100	60	126.2	0.0	0.0	180.3	2.32
9 Jul	210	Mid	0.0	1.00	100	58	121.4	0.0	0.0	173.4	2.23
25 Jul	226	End	0.0	1.00	100	75	157.5	0.0	0.0	225.0	1.63
8 Aug	End	End	0.0	1.00	0	32					

Appendix C-1: Irrigation Schedule table



Appendix C-2: Irrigation Schedule Chart